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News media analytics in finance: a survey

Tom Marty[a], Bruce Vanstone[a], Tobias Hahn[b]

[a]*Bond University, Gold Coast*
[b]*Dayap Logic Pty Ltd, Milton, QLD, Australia*

Abstract

We survey the works applying text analytics to the study of news media in financial markets beyond intraday horizons, and expand into the fundamental economic theory and concepts relevant to the field. We compare and contrast the news sources, textual analysis methods and empirical modelling approaches adopted within the literature. We distil and categorise the key empirical insights, and summarise the bibliographic history of the literature so far. While this rapidly growing field has yielded many exciting discoveries, there are a number of promising avenues for future research which will only benefit from continued advances in computational technology.

Key words: Media coverage; Textual analysis; News tone; Investor sentiment; Sentiment analysis

JEL classification: D82, D83, G14, G12, G41

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1. Introduction

All market prices are reflections of aggregate expected values and the corresponding supply–demand equilibria. As such, any source of information on which investors routinely condition their expectations may plausibly be linked to market behaviour. News media, which contains qualitative descriptions of global economic occurrences and firms' production activities, is one

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Please address correspondence to Tom Marty via email: tom.marty@student.bond.edu.au

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such source of information. Researchers and practitioners alike have long recognised that the financial press plays an influential role in the markets, but until relatively recently, analyses involving news content have not been scalable. It is only within the last two decades, through the widespread digitisation of text, vast expansion of online media resources and the evolution of big data technologies, that quantifying large sets of unstructured text has become feasible. In particular, recent advances in natural language processing and information retrieval capabilities have presented a new dimension of inquiry, text analytics, through which to examine the markets.

In this paper, we provide a systematic survey of the growing literature applying text analytics to the study of news media in financial markets. Although this article is relevant to the use of text analytics in finance more generally, in characterising the empirical research we focus exclusively on investigations for which news media is the predominant textual medium and on market dynamics beyond intraday horizons. The latter condition arises for two main reasons. First, much of the intraday literature is concentrated on the task of machine learning prediction, which has been the major focus of several prior surveys of the news analytics literature (Nassirtoussi *et al*., 2014; Nardo *et al*., 2016). Second, intraday news-market dynamics and the strategies presented to exploit them are likely to be unsuitable for many high-capacity fund managers bound to lower frequency fund rebalancing. By excluding intraday literature from our sample, we focus on the comparatively neglected research examining the low-frequency news dynamics relevant to market valuations.

A number of excellent papers have already summarised various parts of the news analytics literature, yet each of these cover different collections of publications and focus on different aspects of the field compared to this article. Unlike previous surveys, we focus exclusively on news media as opposed to other sources of text used in the literature, and cover a unique cross section of theory and empirical insights relevant to both academics and practitioners. As noted by Loughran and McDonald (2016), any attempt to distil research in this area quickly becomes dated due to the increasing use of textual analysis in finance and accounting. In this respect, we provide an updated contribution to the survey literature.

Notable survey publications include Li (2010), Kearney and Liu (2014), Nassirtoussi *et al*. (2014), Tetlock (2014), Loughran and McDonald (2016) and Nardo *et al*. (2016). The monograph by Das (2014) and the news analytics handbook by Mitra and Mitra (2011) are also useful resources for those interested in the field.

The remaining sections are organised as follows. Section 2 presents the main theoretical models and concepts that set the stage for the study of news media in finance. As these models describe mechanisms by which public information arrival can influence the formation and behaviour of asset prices, they provide a strong basis from which to interpret and develop the empirical research. Here, we also consider news media as a source of textual information and discuss

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some of the properties which make it an attractive medium for the application of textual analysis in finance.

Following Hagenau *et al.* (2013) and Nassirtoussi *et al.* (2014), we discuss textual analysis methodology in terms of the four main stages involved in quantifying textual information: feature extraction, feature selection, feature representation, and document scoring and classification. In Section 3, we compare the core textual analysis techniques used to transform raw financial text into the numeric data used for modelling and testing, and how the different approaches are represented in the literature. Section 4 discusses the broad categories of methods used for empirical analysis and hypothesis testing, and introduces the main approaches to news-based variable construction and aggregation. Section 5 summarises the main findings of the literature to date. Section 6 concludes the paper and suggests avenues for future research. A list containing a summary of each of the reviewed publications is provided as an Appendix.

2. Theoretical background

Theoretical models provide a basis for understanding the potential causal factors underpinning the themes and findings of the news media analytics literature, and offer researchers a credible starting point for constraining the empirical specifications of their analyses in the face of expansive contemporary data sets. Tetlock (2007), Li (2010), and Loughran and McDonald (2016) each cite the importance of economic theory in understanding and progressing text-driven research in finance.

Every trade requires willing participants on either side. Speculative markets with only rational and informationally informed trading result in conditions of adverse selection in which participants are no longer incentivised to trade, and so the market ceases to function (Akerlof, 1970; Milgrom and Stokey, 1982). Noise trading (non-informational trading) supplies the liquidity required for informed trading to exist and thereby carries genuine information into prices, whilst ensuring price is only a noisy reflection of value. This partially revealing property of price is required to incentivise the collection of information (Grossman and Stiglitz, 1976). Thus, information asymmetry and non-informational trading are inextricable properties of financial markets.

Noisy rational expectations (RE) models (Kyle, 1985) provide a rich theoretical framework in which to examine these features of markets and how they may interact in response to public information arrival from sources such as news media. Broadly, they examine how agents with a decisive information advantage will strategise their trading to minimise the price response and information decay of their orders, and how the uninformed will trade in the presence of adverse selection and public information arrival. This depends on the incidence (Wang, 1994; Tetlock, 2010), quality (Foster and Viswanathan, 1990) and unexpectedness (Foster and Viswanathan, 1993b) of

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public information, the behaviour of liquidity traders (Admati and Pfleiderer, 1988), the number of informed (Holden and Subrahmanyam, 1992) and the similarity in information among the informed (Foster and Viswanathan, 1993a). Other formulations of RE discard the notion of a strategic information advantage and consider situations in which private information takes the form of differential information between investors (He and Wang, 1995).

In relation to news, such models suggest that public information will generally act to resolve information asymmetry, leading to a convergence of beliefs (Foster and Viswanathan, 1990; He and Wang, 1995; Tetlock, 2010). Commonly observed signals will be rapidly incorporated into prices (Holden and Subrahmanyam, 1992) but those with superior information, or information processing capability can identify profitable trading opportunities (Admati and Pfleiderer, 1988; Foster and Viswanathan, 1993a). Traders with a long-lived private informational advantage will trade on their information over multiple periods and at an intensity proportional to the quality of public signals (Foster and Viswanathan, 1990). Variations in the degree to which private information is correlated, and the unexpectedness of public information events influence the relationship between price, volume and news flow (Foster and Viswanathan, 1993a). Collectively, these findings provide a rational basis for considering news-driven trading reactions at various horizons and in different information environments (see, for example, Engle *et al*., 2012).

A traditional view of asset pricing holds that noise traders (delegated the role of liquidity shocks in the above models) cannot meaningfully impact prices, as they will be traded against aggressively by rational arbitrageurs who will drive prices close to fundamentals in the process. Black (1986), however, argued that noise-induced deviations from fundamentals may persist through time due to the capital constraints and risk aversion of rational investors. DeLong *et al.* (1990) and Campbell and Kyle (1993), in the spirit of Black (1986), developed equilibria in which random innovations in noise trader bias (i.e. sentiment) can have a significant and prolonged bearing on asset return and trading volumes. As demonstrated by Tetlock (2007), such theories of investor sentiment lead to testable predictions if news content is taken to reflect investor sentiment. The theoretical link between noise trader behaviour and public information is made explicit by difference of opinion (DO) models.

In contrast to RE models, investors in DO models have heterogeneous priors and interpret information differentially, allowing them to 'agree to disagree' (Kandel and Pearson, 1995; Banerjee and Kremer, 2010). Hence, noise trading need not remain an exogenous phenomenon, but instead may be provoked through misinterpretation of valid information. This notion is invoked heavily within the news analytics literature, such that measures of 'textual sentiment' are held not only as an empirical proxy for investor sentiment, but as an influencer of it.

Behavioural finance models motivated by specific cognitive biases and bounded rationalities have sought to explain empirical anomalies such as stock

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price momentum (Jegadeesh and Titman, 1993), event-based return predictability, and long-term reversals (De Bondt and Thaler, 1985) through investors' interaction with different types of public information. For example, the theories of Barberis *et al.* (1998), Daniel *et al.* (1998), and Hong and Stein (1999) have featured prominently in the low-frequency news analytics literature. These models are relevant to the sequential release of information, the gradual diffusion of information across different investor groups, the joint effects of endogenous and exogenous information, and cross-sectional implications following the interaction between media exposure and market expectations.

An alternative perspective sees the role of investor attention and media exposure on stock valuations more directly, whereby firms with low news exposure are relatively unknown so must offer their investors higher returns as compensation for imperfect diversification (Merton, 1987) or for information asymmetry in which a higher degree of firm information remains private (Easley and O'Hara, 2004). Behavioural models such as those of Hirshleifer and Teoh (2003) and Peng and Xiong (2006), which emphasise the role of investors' limited attention towards different types of news content, also imply cross-sectional effects related to firms' information environments and news exposure.

An often-cited theoretical basis for the influence of news on asset variance is the Mixture-of-Distributions Hypothesis (Clark, 1973; Epps and Epps, 1976; Tauchen and Pitts, 1983), which posits that volatility and volume are mixtures of independent distributions for which the rate of information arrival is the mixing variable. As discussed by Engle *et al.* (2012), this is a general hypothesis capable of subsuming many of the more detailed hypotheses relating information arrival and transmission characteristics to the behaviour of return volatility and trading volume.

2.1. Financial news as a source of public information

Textual analysis is applied in the accounting and finance literature to capture the opinions, emotion and value-relevant information embedded within linguistic content. In this sense, the type of information intended to be captured is in large part reflected by the source of the textual data used for the analysis, with most finance research focusing on one of three main sources: corporate disclosures and filings, news media and user-generated content (UGC) such as social media.

Each of these sources has their relative advantages and disadvantages for different types of analysis, but the unique characteristics of news content make it a particularly flexible medium for textual analysis. Unlike corporate disclosures, news media is produced by a large number of sources, covers a diverse range of entities and events, and offers the timeliness, frequency and volume of content afforded by a 24-7 press cycle. And in contrast to UGC,

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which may be interpreted as the 'mood' of the retail investor (Bollen *et al.*, 2011; Li *et al.*, 2014a), various channels of the financial press are widely recognised as credible sources of information relevant to fundamentals. This means that sentiment conveyed in news media may be relevant to the market dynamics of a wider range of securities compared to UGC, for which the effects may be concentrated among small market capitalisation securities (Leung and Ton, 2015).

These aspects allow researchers to study the impacts of news content aggregated over varying scales of time and entity. For example, researchers have studied the influence of news at the firm level (Tetlock *et al.*, 2008; Ferguson *et al.*, 2015), industry level (Li *et al.*, 2014b; Smales, 2015b), market level (Tetlock, 2007; Wei *et al.*, 2017), commodity level (Clements and Todorova, 2016; Maslyuk-Escobedo *et al.*, 2017), between currencies (Nassirtoussi *et al.*, 2015), across countries (Griffin *et al.*, 2011) and at the global market level (Uhl *et al.*, 2015). Time horizons vary from intraday (Groß-Klußmann and Hautsch, 2011; Ho *et al.*, 2013), daily (Tetlock, 2011; Garcia, 2013), weekly (Lu and Wei, 2013; Sinha, 2016), monthly (Ammann *et al.*, 2014; Cahan *et al.*, 2017) and over several years (Hillert *et al.*, 2014; Kraussl and Mirgorodskaya, 2017).

News content also enables analysis over different scales of information richness. The literature tends to differentiate between subcategories of news with respect to their information content, with equivalent interpretations of the analytics derived from them. Intraday newswires appear to provide more timely and value-relevant information than lower frequency newspapers (Tetlock *et al.*, 2008; Ahmad *et al.*, 2016), and differences in firm coverage between newspapers can result in measurable differences in market response (Ferguson *et al.*, 2015). Stories comprising a number of smaller takes or those explicitly related to fundamentals also appear to offer an increased informational advantage (Tetlock *et al.*, 2008; Tetlock, 2010).

Other sources of financial reporting, such as *post mortem* market commentary, enable researchers to investigate the impact of news content in a setting where value-relevant information is less likely to drive results (Tetlock, 2007; Garcia, 2013). Efforts to completely remove informational and market-endogenous aspects have been met using exogenous drivers of news content such as journalist scheduling (Dougal *et al.*, 2012) and differences between media outlets (Engelberg and Parsons, 2011).

The informational characteristics of news variants are also of concern to practitioners, insofar as it affects the ability to explain and forecast market behaviour. For example, stock prices respond less to stale news but the extent of overreaction, or short-term reversal, is greater (Tetlock, 2011). To this end, removing news items containing old information can increase the performance of high-frequency news-based signals (Leinweber and Sisk, 2011). However, what is considered stale in the short term is often the result of greater journalist attention or unfolding news events. This makes repeated or 'stale' news a

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potential proxy for important and complicated events that are gradually processed by investors and therefore a useful measure for low-frequency strategies (Huynh and Smith, 2017).

2.2. Publication selection

To establish a baseline citation corpus, a systematic search was performed on the Clarivate Analytics Social Sciences Citation Index (SSCI) from 1900 to 2018 using the Web of Science platform. The SSCI was queried for all peer-reviewed articles with titles, abstracts and keywords fulfilling any one, or combination of, the following search criteria:

TS = (((sentiment OR tone OR analytics) AND (news OR media)) AND ((stock OR market) AND (volatility OR risk)))
TS = ((((language OR linguistic) AND (content OR tone OR positive OR negative)) AND (news OR media)) AND ((stock OR market) AND (volatility OR risk)))
TS = ((((language OR linguistic) AND (content OR tone OR positive OR negative)) AND (news OR media)) AND ('stock market' OR 'stock return*' OR 'stock price' OR volatility))
TS = (((sentiment OR tone OR analytics) AND (news OR media)) AND ('stock market' OR 'stock return*' OR 'stock price' OR volatility))
TS = ((news OR media) AND (sentiment OR tone OR analytics) AND ('trading strategy'))

This returned 276 items, to which an additional 10 items (Wüthrich *et al.*, 1998; Chan, 2003; Kothari *et al.*, 2009; Mitra *et al.*, 2009; Tetlock, 2010; Leinweber and Sisk, 2011; Ferguson *et al.*, 2015; Yang *et al.*, 2015; Sinha, 2016; Manela and Moreira, 2017) were added from other sources, such as Google Scholar. To remove irrelevant items and establish a refined corpus subject to individual written review, the 286 initial items were filtered by removing those that:

1 Did not make use of measures derived from financial news content.
2 Did not examine forward market movements at time horizons of at least one day.
3 Did not empirically examine movement of market-based variables.
4 Only examined market responses surrounding earnings announcements.
5 Studied only a particular news event (case studies) or events (e.g. airline crashes) or a particular type of stock (e.g. Islamic stocks).

Two exceptions were made to rule number one, for publications in which content-derived measures were not used – Chan (2003) and Fang and Peress (2009) – as these two studies have been particularly influential in the

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low-frequency literature. The filtering criteria excluded a total of 236 items, leaving 50 items, exceptions included, for review.

3. Textual analysis techniques

3.1. Feature extraction

Document representation, or feature extraction, refers to the early parsing procedure in which documents are broken down into the constituents that will be used in subsequent processing steps. This is a fundamental step in the textual analysis pipeline as it determines the type and amount of the original information available for subsequent analysis.

Most approaches to textual sentiment require analysis beyond the character level. As such, the running text must first be segmented into meaningful units (or tokens), with words being the common choice of unit. This process of identifying words (or equivalently, their boundaries) is referred to as *tokenisation* or *word segmentation*. For many approaches to sentiment analysis, word segmentation is the only form of feature extraction required, allowing document representation to remain at the word level. This is embodied by the most common method of document representation used in textual analysis – the bag of words (BOW) approach – in which the document corpus is collapsed into a *term-document matrix*. The critical assumption underlying the BOW approach is term exchangeability. As shown in Table 1, BOW is used in the majority of the review items not using pre-processed data.

Common extensions to the BOW representation include *N-grams* and *skip-grams*. *N*-grams consist of continuous sequences of *n* words, while skip-grams are *n*-word combinations allowing for a non-zero distance between words. As *N*-grams and word combinations retain some sequencing information, they offer a compromise between the generic BOW and higher-level string-based representations.

In moving beyond the word level, a natural requirement is the identification of sentences from running text, known as *sentence tokenisation* or *sentence segmentation*. Sentences represent structurally independent grammatical units and are the subject of most grammars (Grefenstette and Tapanainen, 1994). Sentences therefore form the basis for higher-level natural language processing tasks that aim to abstract meaning using grammars (Section 3.3.2). Part-of-speech (POS) tagging is a fundamental step in this abstraction process. It aims to label each word with a tag indicating its appropriate part-of-speech (word class or syntactic category). Parts of speech provide a large amount of information about the meaning of a sentence, and certain parts of speech, such as adjectives, are particularly important indicators of opinions (Liu and Zhang, 2012). POS tagging is also a means of identifying the occurrence of negation and intensifiers, which flip and strengthen the meaning of other words, respectively.

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Table 1
Custom textual analysis techniques

Author	Text source	Feature extraction	Feature selection	Feature representation	Dictionary
Chan and Chong (2017)	News articles and blogs from Finet website	Words, sentences	POS tags, term weights, tree topological features, dictionary	Parse tree	H-IV4, SentiWordNet
Kraussl and Mirgorodskaya (2017)	*Wall Street Journal*, *New York Times*, *Financial Times*	BOW	Dictionary	Term occurrence	Custom variation of LM
Manela and Moreira (2017)	*Wall Street Journal*	Unigrams, bigrams	Stopword removal	Term frequency	NA
Narayan and Bannigidadmath (2017)	*New York Times*	BOW	Dictionary	Term frequency	LM
Seng and Yang (2017)	KMW database – *China Times*	BOW	Dictionary, Chi-square	Term frequency	Custom
Wei *et al*. (2017)	*China Times* and *Commercial Times* from InfoTimes database	BOW	Dictionary	Term frequency	Dictionary (Chinese translation)
Yang *et al*. (2017)	Northern Light Single Point online business news portal	BOW	Dictionary, lemmatisation	Term weighting (Dictionary)	SentiWordNet
Wu and Lin (2017)	Market Observation Post System, and newspapers and websites from *Taiwan Economic Journal*	BOW	Dictionary	Term occurrence	Lu and Wei (2013) Chinese word list
Ahmad *et al*. (2016)	Newspapers, industry and trade magazines, newswires, financial blogs, web-based publications from LexisNexis database	BOW	Dictionary	Term frequency	LM

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Table 1 (continued)

Author	Text source	Feature extraction	Feature selection	Feature representation	Dictionary
Zhang *et al.* (2016a)	Articles and UGC from NASDAQ website	BOW with simple negation	Stemming (Porter), Dictionary	Term frequency	MPQA (Wilson *et al.*, 2005), LM, BL (Hu and Liu, 2004)
Strauß *et al.* (2016)	Five leading Netherlands newspapers	BOW	Dictionary	Term frequency	LIWC
Kroujiline *et al.* (2016)	Factiva database, online news sources	Not provided	Phrase lexicon	Term occurrence	Custom
Zhang *et al.* (2016b)	NetEase Financial Channel (Chinese website)	Occurrence only	NA	NA	NA
Yang *et al.* (2015)	Northern Light Single Point online news portal	BOW	Dictionary, lemmatisation	Term weighting (Dictionary)	SentiWordNet
Wang *et al.* (2015)	CMoney database	BOW	Dictionary	Not reported	Not reported
Ferguson *et al.* (2015)	*Financial Times*, *Guardian*, *Mirror* from LexisNexis UK database	BOW	Dictionary	Term frequency	LM
Bianconi *et al.* (2015)	*New York Times*	BOW	Dictionary	Term frequency	Custom variation of LM
Ammann *et al.* (2014)	*Handelsblatt* (German newspaper)	BOW	Dictionary	Term occurrence	LM
Hillert *et al.* (2014)	*New York Times*, *USA Today*, *Wall Street Journal* + 41 local newspapers, from LexisNexis database	BOW	Dictionary	Term frequency	LM
Li *et al.* (2014b)	Macroeconomic and company news from FINET archive	BOW	Dictionary	Term weighting (Dictionary)	H-IV4, LM, SenticNet
Yu *et al.* (2013)	Google Blogs, BoardReader, Twitter, Google News – ABC	BOW	Not stated	Term frequency	NA

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Table 1 (continued)

Author	Text source	Feature extraction	Feature selection	Feature representation	Dictionary
	News, *New York Times*, *USA Today*, Fox News, Reuters, *Wall Street Journal*, *Washington Post*, CNN, *The Economist* and Forbes.				
Garcia (2013)	*New York Times*	BOW	Dictionary	Term frequency	LM
Xiong and Bharadwaj (2013)	Lydia/TextMap (~ 1000 online news sources)	Words, sentences	NER, POS tags, synonym set, dictionary	Term frequency	Lydia/TextMap
Dougal *et al.* (2012)	*Wall Street Journal*	Journalist, BOW	Dictionary	Term frequency	LM
Tetlock (2011)	Dow Jones newswires from Factiva database	Unigrams, bigrams	Not stated	Term occurrence	NA
Tetlock (2010)	Dow Jones newswires from Factiva database	BOW	Keyword	Term occurrence	NA
Fang and Peress (2009)	*New York Times*, *USA Today*, *Wall Street Journal* and *Washington Post* from the LexisNexis database	Item occurrence	NA	NA	NA
Kothari *et al.* (2009)	EDGAR, Factiva, Dow Jones Interactive	BOW	Dictionary	Term frequency, term occurrence	Custom, H-IV4
Tetlock *et al.* (2008)	*Wall Street Journal*, Dow Jones newswires	BOW	Dictionary	Term frequency	H-IV4
Tetlock (2007)	*Wall Street Journal*	BOW	Dictionary	Term frequency projected onto first principal component	H-IV4

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Table 1 (continued)

Author	Text source	Feature extraction	Feature selection	Feature representation	Dictionary
Chan (2003)	Dow Jones Interactive Publications Library – *Wall Street Journal*, *New York Times*, *LA Times*, *Globe*, *Chicago Tribune*, etc.	Item occurrence/ stock direction	NA	NA	NA
Wüthrich *et al.* (1998)	*Wall Street Journal*, *Financial Times*, Reuters, Dow Jones and Bloomberg	Unigrams to 5-g	Phrase lexicon	Term weighting	Custom

BOW, bag of words; POS, part of speech; H-IV4, Harvard IV-4; LM, Loughran and McDonald (2011) word list.

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A named entity in a piece of text is a sequence of words that designates a real-world entity, such as a person, organisation or location. Named entity recognition (NER) refers to the combined task of identifying and classifying such entities. In its basic form, NER provides a means of matching news articles to entities. In more advanced systems, NER allows attribution of particular portions of text within an article to the entities being discussed. NER can also be used for estimating the relevance of a news item to a given entity (Thomson Reuters, 2013), as an additional feature set in itself (Schumaker and Chen, 2009), or for event identification and information extraction (Boudoukh *et al.*, 2013).

3.2. Feature selection

Feature selection is used to determine the features which retain the most relevant information for the classification task and to remove redundant data. It is of particular importance in textual analysis applications due to the high dimensionality and noisiness of word-based representations.

Two of the most common forms of feature selection, which are generally used in addition to other techniques, are stop word removal and morphological stemming (lemmatisation). Stop word removal simply removes words that are extremely common but unlikely to provide specific or discriminatory information. Stemming and lemmatisation aim to reduce inflectional and derivative forms related to a common base form.

Beyond these standard means of dimensionality reduction, an important and prevalent approach in the finance literature has been the use of dictionary-based feature extraction. This approach typically filters features based on whether they occur within predefined lexicons (word lists) which have been manually identified by domain experts to capture particular elements of the text. The resulting feature space is a term-document matrix containing the incidence or frequency of the lexicon terms.

The most frequently used word lists in the accounting and finance literature are the Henry (2008) word list, the General Inquirer (GI) Harvard IV-4 (H-IV4) word lists (Stone *et al.*, 1966), the Diction (Hart, 2000) word lists, and the Loughran and McDonald (2011) (LM) word lists. Diction and H-IV4 are general linguistic word lists premised on sociology and psychology and were not developed for use within any one specific domain. The Henry (2008) and LM word lists were designed specifically for finance and accounting, and were based on the content of earnings press releases and 10-Ks, respectively. In general, finance-specific word lists are more suitable for financial applications; they are based on the most likely interpretation of words in a business context, which may differ substantially from their everyday connotations, and they tend to correlate more strongly with variables of interest (Loughran and McDonald, 2011, 2016).

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An alternative to using a predefined lexicon is to perform feature selection based on the statistical properties of the document corpus itself. This may be as simple as discarding terms with an occurrence rate below a given threshold, but typically involves the use of statistical weighting schemes that aim to measure the semantic focus or class discriminatory power of each term. Common weighting schemes include term-frequency-inverse-document-frequency (TF-IDF), information gain or entropy, Fisher's discriminant (Fisher, 1936), the Gini index, mutual information and the Chi-square statistic. A brief definition and overview of these measures is provided by Aggarwal and Zhai (2012).

3.3. Feature transformation and representation

After feature selection has reduced the original set of attributes to a smaller, information-rich subsample, the remaining features need to be represented numerically for them to be processed by the document scoring and classification procedure. This is known as *feature representation*. Representation of the reduced feature set is often the same as the original set (e.g. a standardised vector of term frequencies), or comprises the probabilistic term scores used for feature selection. The latter may be implemented by weighting the occurrence and frequency values in the term-document vector by the relevant score for each term. In the case of dictionary-based feature selection, each term score may also be weighted as positive or negative according to its category.

Feature representation may also involve creating a new (and smaller) set of features, i.e. *feature transformation*.[1] Examples include Latent Semantic Indexing (LSI) (Tetlock, 2007), hierarchical clustering (Ammann *et al.*, 2014), probabilistic LSI, latent Dirichlet allocation and linear discriminant analysis. See Lee *et al.* (2010), Aggarwal and Zhai (2012) and Crain *et al.* (2012) for an overview of these and other feature transformation techniques.

3.4. Document scoring and classification

Once features have been extracted, selected, transformed and represented in a machine-readable format, document classification and scoring is conducted. The term 'document' here simply refers to the collection of terms used to create the score and may include multiple news items.

There are two general ways this is performed in the literature – analytically, or via machine learning. Analytical scoring is frequently used in conjunction with dictionary-based measures, and typically applies one or more of the following metrics: the sum of positively associated terms, the sum of negatively associated terms, and the total sum of terms (if the feature representation was not normalised to document length). Each of these summations will have

[1] Whether or not this is considered a feature selection or feature representation process varies from author to author.

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slightly different meanings depending on what feature representation was selected.

For statistical scoring, the same approach can be taken as with any machine learning classification procedure – at this point there is nothing special about the fact that the vector values are representations of some textual attribute. Common techniques include neural networks (Thomson Reuters, 2013), Bayes classifier (Shi *et al.*, 2016), support vector machines (SVM), decision trees, random forests and k-nearest neighbours (kNN). Reviews of these and other classification methods, and their relative performance in the context of text mining, are provided by Sebastiani (2002) and Ravi and Ravi (2015).

An important distinction for hypothesis formation and testing is whether the classification training corpus is labelled from exogenous or endogenous sources; this also applies to the probabilistic term weighting schemes discussed above. This is generally reflected in the literature as the difference between whether documents were labelled manually by a human (Antweiler and Frank, 2004; Thomson Reuters, 2013) or on market behaviour surrounding the document's publication (Hagenau *et al.*, 2013; Li *et al.*, 2014b; Manela and Moreira, 2017).

3.5. Grammar-based systems

The feature selection and transformation methods described above are more easily understood in terms of word or *n*-gram term-document matrix representations than the more advanced sentence-level representations used by state-of-the-art natural language processing (NLP) systems. These systems apply formal models of language structure such as context-free grammar (CFG) or dependency grammars to build a parse tree representation of the text that reflects the relationships between words and phrases. Modified versions of the extraction and scoring techniques already discussed are also utilised, but in a hierarchical procedure that leverages the linguistic structures identified by the grammatical parsing. This type of textual processing resembles that used within proprietary news analytics feeds (Table 2) such as those provided by Thomson Reuters, and by Stanford's CoreNLP (Thomson Reuters, 2013; Stanford NLP Group, 2018). A custom grammar-based system is also presented by Chan and Chong (2017).

4. Econometric techniques

Each finance study making use of news analytics consists of two broad components: attainment of features derived from the arrival and content of news text, and econometric analysis and hypothesis testing. While the textual analysis techniques and the econometric models employed for a given study should be consistent with the overall hypothesis, the two components can generally be considered independently. Given this, there are a number of ways

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Table 2
Use of proprietary textual analysis packages

Author	Text source	Package/Database	Variables used
Cahan *et al.* (2017)	Company news	TRNA	Sentiment probability (positive, negative)
Heston and Sinha (2017)	Company news	TRNA	Sentiment probability (positive, negative, neutral), novelty
Huynh and Smith (2017)	Company news	TRNA	Sentiment probability (positive, negative), relevance, novelty, headline
Borovkova and Mahakena (2015)	Macroeconomic news (natural gas)	TRNA	Sentiment probability (positive, negative, neutral), novelty
Allen *et al.* (2017)	Company news	TRNA	Sentiment probability (positive, negative)
Song *et al.* (2017)	Company news	TRNA	Sentiment probability (positive, negative), relevance
Smales (2016)	Company news	RavenPack	MCQ sentiment score, relevance, novelty
Nooijen and Broda (2016)	Online company news and social media	TRMI	Social media and news sentiment indices
Shi *et al.* (2016)	Company news	RavenPack	Relevance, novelty, CSS, ESS
Sinha (2016)	Company news	TRNA	Sentiment probability (positive, negative), novelty, relevance
Khuu *et al.* (2016)	Market and company news	TRNA	Sentiment probability (positive, negative), novelty, relevance
Hendershott *et al.* (2015)	Company news	TRNA	Sentiment probability (positive, negative), relevance, topic codes
Smales (2015a)	Macroeconomic news (gold)	TRNA	Sentiment probability (positive, negative), novelty, relevance
Smales (2015b)	Industry and company news	TRNA	Sentiment probability (positive, negative), novelty, relevance
Uhl *et al.* (2015)	Company and macroeconomic news	TRNA	Sentiment probability (positive, negative)
Nuij *et al.* (2014)	Company news	ViewerPro	Event classification, event impact score
Uhl (2014)	Market and company news	TRNA	Sentiment classification
Mitra *et al.* (2009)	Company news	RavenPack	Relevance, ESS

MCQ, CSS and ESS represent RavenPack's *multiclassifier for equities*, *composite sentiment score*, and *event sentiment score*, respectively (Mitra and Mitra, 2011).

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presented in the literature in moving from the attainment of news-based features to econometric testing. In the context of this discussion, econometric testing broadly refers to methods of predicting or explaining market variables in terms of news variables. The econometric techniques used within the reviewed works are presented in Table 3.

4.1. Parametric techniques

In a parameter-focused approach, the testing of hypotheses concerning the relationship between news flow and market dynamics is largely reduced to an estimation problem. Parametric methods first assume a functional form of the relationship between the predictors and response being examined. This functional form is described explicitly using a model which is then fitted to the data by estimating the set of model parameters according to some objective function.

The nature of the relationship between the predictors and response can then be judged on the basis of the statistical significance and magnitude of the model parameters and model closeness-of-fit statistics. Simple parametric models are relatively inflexible and are likely to mischaracterise the true functional relationship between the predictors and response. However, they are highly interpretable and therefore suited to hypotheses requiring inference as to the role of individual predictors in the modelled system.

4.1.1. Linear regression

The most common parametric approach in the news analytics literature is multiple linear regression – a single equation linear model with multiple predictors, fit using ordinary or generalised least squares. Typically, the dependent variable used is raw return or abnormal return calculated from the Fama–French three-factor model or Capital Asset Pricing Model (CAPM), and the independent variables include current and lagged values of the relevant news measures and control variables. Autocorrelation and heteroscedasticity in the residuals are commonly corrected using Newey and West (1987) or White (1980) standard errors.

Of the relevant studies considered for this review, multiple linear regression is used as the primary econometric technique by Tetlock *et al.* (2008), Kothari *et al.* (2009), Dougal *et al.* (2012), Garcia (2013), Lu and Wei (2013), Yu *et al.* (2013), Ammann *et al.* (2014), Bianconi *et al.* (2015), Ferguson *et al.* (2015), Hendershott *et al.* (2015), Smales (2015b), Yang *et al.* (2015), Khuu *et al.* (2016), Shi *et al.* (2016), Strauß *et al.* (2016), Zhang *et al.* (2016a), Cahan *et al.* (2017), Narayan and Bannigidadmath (2017), Seng and Yang (2017), and Wu and Lin (2017). Logistic regression is used when the response variable is categorical, and the objective is to model the posterior probabilities of each class as linear functions of the inputs. Hendershott *et al.* (2015) use logistic

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Table 3
Use of econometric techniques

Author	Market variable	Estimation horizon	Econometric technique
Chan and Chong (2017)	Return	5 days	Granger causality
Manela and Moreira (2017)	Return, volatility, VIX	1–24 months	SVR, linear regression
Seng and Yang (2017)	Volatility	1 year	Linear regression forecasts
Yang *et al.* (2017)	Return	≈1 day	Genetic algorithm, trading simulation optimisation
Huynh and Smith (2017)	Return	1 year	Portfolio analysis
Song *et al.* (2017)	Return	1 week	Neural network, trading simulation
Cahan *et al.* (2017)	Volatility, liquidity volatility	1 month	Linear regression
Wei *et al.* (2017)	Return, trading value, turnover ratio, volatility	1 week, 1 month	VAR, portfolio analysis
Heston and Sinha (2017)	Return	13 weeks	Portfolio analysis, event study
Kraussl and Mirgorodskaya (2017)	Return, volatility	1 month	VAR, Granger causality
Wu and Lin (2017)	Return, institutional order flow	1 month contemporaneous	Linear regression
Borovkova and Mahakena (2015)	Volatility, returns	1 day	GARCH, HEAVY, GJR-GARCH, VAR, Granger causality
Allen *et al.* (2017)	Returns	1 day	Linear regression, EGARCH, GJR-GARCH, entropy, MI
Narayan and Bannigidadmath (2017)	Return	1 day	Linear regression, trading simulation
Smales (2016)	VIX, return	Contemporaneous, 1 –250 days	VAR, linear regression
Nooijen and Broda (2016)	Volatility, return	1 day	EGARCH, MS-GARCH
Shi *et al.* (2016)	Volatility, return	1 month	EGARCH, linear regression
Ahmad *et al.* (2016)	Return	Up to 250 days	VAR
Zhang *et al.* (2016a)		1 day	VAR

(continued)

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Table 3 (continued)

Author	Market variable	Estimation horizon	Econometric technique
	Volume, volatility, return		
Sinha (2016)	Return	13 weeks	Portfolio analysis, trading simulation
Strauß *et al.* (2016)	Return	1 day	VAR, Granger causality
Kroujiline *et al.* (2016)	Return	≈45 days	Trading simulation, model fitting
Zhang *et al.* (2016b)	Return, volume	50 days	Event study, linear regression
Khuu *et al.* (2016)	Return	1 day contemporaneous	Linear regression
Smales (2015a)	Volatility	1 day	Linear regression, TGARCH
Smales (2015b)	Return	Contemporaneous	Linear regression
Uhl *et al.* (2015)	Return	Dynamic, 1 week	Trading simulation, linear regression
Yang *et al.* (2015)	Return, volume, volatility	11 days	Linear regression
Wang *et al.* (2015)	Return	5 days	VAR, Granger causality, impulse response
Ferguson *et al.* (2015)	Return	1 day	Linear regression, trading simulation
Bianconi *et al.* (2015)	Volatility, Co-volatility, VIX	Contemporaneous, 1 day	Linear regression, Granger causality
Hendershott *et al.* (2015)	Return, institutional order flow	10 days	Logit regression, event study, linear regression, VAR, impulse response
Ammann *et al.* (2014)	Return, industrial production	1 month, 3 months	Linear regression
Li *et al.* (2014b)	Return	1 day	Support vector machines
Uhl (2014)	Return	1 month	VAR, trading simulation
Hillert *et al.* (2014)	Return	6–36 months	Portfolio analysis, event study
Nuij *et al.* (2014)	Return	1, 3, 5 days	Genetic algorithm, trading simulation optimisation
Xiong and Bharadwaj (2013)	Return	1 month	Dynamic panel GMM, VAR
Yu *et al.* (2013)	Return, volatility	1 day	Linear regression

(continued)

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Table 3 (continued)

Author	Market variable	Estimation horizon	Econometric technique
Garcia (2013)	Return	5 days	Linear regression
Dougal *et al.* (2012)	Return	1 day	Linear regression
Tetlock (2011)	Return	5 days	Linear regression
Leinweber and Sisk (2011)	Return	60 days, 20 days	Event study, trading simulation
Tetlock (2010)	Return, volume	10 days	Linear regression
Fang and Peress (2009)	Return	1–12 months	Portfolio analysis
Kothari *et al.* (2009)	Return, volatility, earnings	Contemporaneous, 1 quarter	Linear regression
Mitra *et al.* (2009)	Volatility	1 day	Linear regression, prediction
Tetlock *et al.* (2008)	Return, earnings	10 days, 6 quarters	Linear regression, event study, trading simulation
Tetlock (2007)	Return, volume	5 days	VAR, event study, trading simulation
Chan (2003)	Return	1–36 Months	Portfolio analysis
Wüthrich *et al.* (1998)	Return	1 day	Rule-based classifier, kNN, ANN, linear regression

Estimation horizons listed with approximate values reflect studies in which the estimation horizon was not fixed but could be estimated, such as dynamic trading signals with a cited average holding period. VIX, volatility index

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regression to estimate the probability of a news event in terms of institutional order volume and the prior day's news incidence.

4.1.2. Vector autoregression

A variant of the single equation regression model is the vector autoregression (VAR) model, which is used in the literature to capture the linear interdependencies between the separate time series of dependent variables, news measures and control variables. That is, the VAR model explains the time series evolution of each variable in terms of their own lagged values and the lagged values of the other model variables. Of the articles considered, VAR models have been used by Tetlock (2007), Uhl (2014), Hendershott *et al.* (2015), Wang *et al.* (2015), Ahmad *et al.* (2016), Smales (2016), Strauß *et al.* (2016), Zhang *et al.* (2016a), Kraussl and Mirgorodskaya (2017) and Wei *et al.* (2017). Xiong and Bharadwaj (2013) used VAR in addition to dynamic panel generalised method of moments (GMM) estimate of the same equation. Properties of the estimated VAR model are often examined through structural analysis techniques such as Granger causality, (and less commonly) impulse responses and forecast error variance decompositions (see Uhl, 2014 for an example of all three). Granger causality is used by Bianconi *et al.* (2015), Wang *et al.* (2015), Strauß *et al.* (2016), Chan and Chong (2017), Checkley *et al.* (2017) and Kraussl and Mirgorodskaya (2017),

4.1.3. Volatility models

Another parametric model commonly used among the studies considered is the GARCH model and its variants. These are used to examine the relationship between news and conditional volatility dynamics whilst allowing for well-known features of financial time series such as volatility persistence, excess kurtosis and distributional asymmetry. Variants of the standard GARCH model used within the literature include EGARCH and GJR-GARCH, which allow for asymmetric response of innovations; Markov-switching GARCH (MS-GARCH), which allows for different volatility regimes; and FI-GARCH, which allows for long memory persistence. Models of the GARCH family are used by Shi *et al.* (2016), Nooijen and Broda (2016), and Allen *et al.* (2017).

4.1.4. Analytic models

Although infrequent in the review sample, the relationship between news information and market variables is sometimes described explicitly through an analytic model. In addition to the clear role of such models in hypothesis formation (see Tetlock, 2010), certain models may be capable of being fit to the data directly. In this case, the model may be used for forecasting, or for

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empirically describing news-market dynamics in ways that were not explicit predictions of the model equations (see Kroujiline *et al*., 2016).

4.2. Nonparametric, semiparametric and model-free techniques

4.2.1. Cross-sectional portfolios

The most prominent nonparametric econometric technique within the reviewed literature is the cross-sectional portfolio approach. In this approach, portfolios reflecting particular portions of cross-sectional input space (consisting of news and market data) measured over a specified formation period are compared in terms of a given response variable (such as abnormal returns) measured over a given holding period. The time, if any, between the formation event and the holding period is termed the skip period, and may be included to avoid capturing microstructure effects or specific market activity such as short-term return reversal. In practice, the conditioning process amounts to ranking all assets based on an input variable and selecting particular groups for portfolio construction (often via a long-short portfolio) and comparison. If more than one variable is being conditioned upon, the ranking and grouping procedure is usually conducted sequentially, with ranking of subsequent variables occurring within each group of the previous variable – this is the case for all portfolio analyses in the review sample conditioning on multiple variables, reflected by the 'Sorting quantiles' column of Table 4.

In the standard portfolio analysis approach, portfolios are formed at every time interval (such as monthly), such that the holding periods overlap and each point in calendar time is associated with the returns of K portfolios, where K is

Table 4
Cross-sectional portfolio analyses

Author	J/S/K	Sorting quantiles	Sorting variables
Song *et al*. (2017)	1W/0W/1W	Decile	ANN output
Wei *et al*. (2017)	1M/0M/1M	Quintile, quintile	Tone, [PB,TO,MV]
Heston and Sinha (2017)	1W/[0,1W]/13W	Decile	Tone
Huynh and Smith (2017)	1W/[0,4W]/52W	Tercile, tercile, tercile	Attention, tone, return
Sinha (2016)	1W/0W/13W	Decile	Tone
Hillert *et al*. (2014)	6M/1M/[6–36M]	Quintile, tercile, median	Coverage, return, tone
Fang and Peress (2009)	[1–6M]/0M/[1–12M]	Binary, median	Incidence, coverage
Chan (2003)	1M/1W/[1–36M]	Binary, tercile	Incidence, return

'J', 'S' and 'K' represent formation, skip and holding period, respectively. 'M' and 'W' reflect months and weeks respectively. 'PB', 'TO' and 'MV' represent price-to-book ratio, turnover and market value, respectively, as defined in Wei *et al*. (2017). Sorting quantiles and their associated ranking variables are listed in the order in which they were performed.

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the length of the formation period (Jegadeesh and Titman, 1993). The equal weighted return series of the overlapping portfolios captures the correlation of returns across event stocks and can be appropriately used to test their average price response (Fama, 1998).

As the output is a time series of returns, the exposure of the portfolio to common risk factors, such as the Fama–French three factors and the Carhart four factors, can be estimated by regressing the return series against the contemporaneous value of the relevant portfolios (market, size, book-to-market and UMD portfolios, for example). To account for the possibility that the documented strategy has nonlinear risk exposures which are not captured by the linear factor models, various authors also report benchmark adjustments (Daniel *et al.*, 1997). Portfolio intrinsic measures such as volatility and Sharpe ratio are also used to indicate risk (Kroujiline *et al.*, 2016).

The cross-sectional portfolio approach is used primarily within the low-frequency news analytics literature and is a particularly useful way of examining the economic significance and exploitability of news-driven pricing effects. The time series of returns produced by the analysis can be interpreted as the returns of a trading strategy implementing the portfolio rules over the test period, while the strategy itself implicitly offers regular and well-defined rebalancing characteristics that are not offered by signal-based strategies. These properties make the technique particularly amenable to portfolio management research. Of the reviewed studies, calendar-time portfolio analysis is used by Chan (2003), Fang and Peress (2009), Hillert *et al.* (2014), Sinha (2016), Huynh and Smith (2017), Song *et al.* (2017) and Wei *et al.* (2017). Heston and Sinha (2017) generate cross-sectional portfolios but do not specify whether results reflect average calendar-time or event-time. Table 4 lists the specifications of the cross-sectional portfolios examined in each of these studies.

4.2.2. Event study

Event studies are another common model-free econometric method often applied in addition to other forms of analysis. Event studies typically attempt to measure the impact of different news events on a firm's stock price or the comparative post-formation performance of different portfolios. Typically, an estimate of the expected or normal value of the variable of interest is determined using an average of recent values or a market model such as the CAPM (in the case of returns). The difference between the normal value of the market variable and its actual value is calculated and plotted throughout an event window featuring the event. Of the review sample, Tetlock *et al.* (2008), Leinweber and Sisk (2011), Hillert *et al.* (2014), Hendershott *et al.* (2015), Zhang *et al.* (2016b), and Heston and Sinha (2017) employed event studies in their analyses.

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4.2.3. Nonparametric models

The parametric econometric techniques applied in the literature are generally linear, in that the response can be expressed in terms of a linear function of the predictors. This imposes restrictions on the relationship between the response and the dependent variables, as well as the interactions between dependent variables, that may not capture the true nature of data in which the statistical relationships are more complex.

In contrast, nonparametric models do not imply explicit *a priori* assumptions about the nature of the functional relationship between the predictors and response. While nonparametric models may contain parameters, the number and nature of the parameters are flexible and determined by the data. Due to their flexibility, nonparametric models are able to estimate a much wider range of news-market functional forms than their commonly used parametric counterparts and are therefore more capable of exploiting inherent features of the data. However, this added flexibility comes at the cost of interpretability, as complicated functional forms obscure the relationship between any individual predictor and the response. For this reason, nonparametric models may be best suited to hypotheses reducible to prediction tasks for which interpretability is not important.

Wüthrich *et al.* (1998) trial kNN and an artificial neural network (ANN) for their daily market prediction system, but decide on a probabilistic rule set. Song *et al.* (2017) use an ANN trained with learning-to-rank loss functions for their trading strategy. Li *et al.* (2014b) use support vector machines (SVM) for their news classification and stock prediction system. Manela and Moreira (2017) use support vector regression (SVR) to predict market implied volatility. The use of nonparametric models within the review sample is limited, suggesting that most hypotheses are primarily associated with understanding the impact of different characteristics of news flow rather than the potential for news analytics to optimise forecasting and trading performance. Performance-focused analyses appear to be more common in the intraday literature (Fung *et al.*, 2003; Schumaker and Chen, 2009; Schumaker *et al.*, 2012; Hagenau *et al.*, 2013; Geva and Zahavi, 2014; Li *et al.*, 2014a; Nassirtoussi *et al.*, 2015; Feuerriegel and Prendinger, 2016; Maragoudakis and Serpanos, 2016).

4.2.4. Forecast comparison

A number of works use the forecasting performance of different (parametric or nonparametric) models, either directly or in a trading strategy, to provide evidence as to the predictive capacity of news content. If the model containing the news-based parameters outperforms the news-free model, this may be interpreted as a demonstration of the predictive content of the news information used. Sometimes the forecasting problem can be interpreted as a classification task (such as directional price predictions); in this case, the

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relative accuracy of the news-driven model may be compared to the accuracy expected by chance (Wüthrich *et al.*, 1998). These types of tests are considered nonparametric because the function relating the news variable(s) to the out-of-sample forecasting performance of models with different specifications is not made explicit. Forecast comparison is used by Ammann *et al.* (2014), Uhl (2014), Bianconi *et al.* (2015) and Nooijen and Broda (2016). This testing philosophy is also the basis for the use of metaheuristics.

4.2.5. Metaheuristics

Metaheuristic procedures are employed to algorithmically select the combination of model type(s), model specifications, inputs and parameters to converge on a combination that delivers a high level of performance in terms a predefined fitness function. As they make few assumptions about the optimisation problem, they are often suitable for complex search spaces that are not well understood, relatively unstructured and too large to be completely sampled (De Jong, 1988, p. 122).

Metaheuristics are used in the news analytics literature as a means to test whether news data, including news-driven events and signals, is a competitive source of conditioning information for predicting stock movements in the context of a trading system. Nuij *et al.* (2014) and Seng and Yang (2017) both employ genetic algorithms (GAs), a metaheuristic belonging to the larger class of evolutionary algorithms, to find optimum trading rules among a set of news-driven and market-driven indicators.

4.2.6. Trading simulations

It is well understood that statistically significant econometric relationships are not necessarily economically meaningful. In order to investigate the economic significance of news-market relationships and the potential relevance to financial practitioners, many researchers have conducted portfolio analyses and trading simulations based on the news information being investigated. News-informed trading strategies are presented by Wüthrich *et al.* (1998), Tetlock (2007), Tetlock *et al.* (2008), Leinweber and Sisk (2011), Nuij *et al.* (2014), Uhl (2014), Ferguson *et al.* (2015), Uhl *et al.* (2015), Yang *et al.* (2015, 2017), Ahmad *et al.* (2016), Kroujiline *et al.* (2016), Smales (2016) and Song *et al.* (2017), in addition to those utilising cross-sectional portfolio analyses (which are directly interpretable as trading strategies).

While such analyses do shed light on the economic size of an effect and help to frame the investigated relationship in terms of its exploitability, there are number of reasons why they may still fail to reflect the performance of a realistic investment strategy. The availability of the assets used for testing, proposed rebalancing frequency, the inclusion of transaction fees, survivorship bias, strategy market depth and reported evaluation metrics are among the

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experimental parameters that determine the economic relevance of a tested strategy.

4.3. Variable construction

Most approaches to econometric testing in the news analytics literature make use of variables constructed from the information contained within multiple news items. It is important to consider this step, as it embeds potentially significant assumptions about the way investors respond to news. When individual items (documents, words or sentences) are scored prior to variable formation, typical measures take the form

$$Pos = \sum_{i}^{N} \frac{(I^{pos} \times S^{pos})}{N} \tag{1}$$

$$Neg = \sum_{i}^{N} \frac{(I^{pos} \times S^{pos})}{N} \tag{2}$$

$$\begin{aligned} Polarity &= Pos - Neg \\ &\quad or \\ &= Neg - Pos \end{aligned} \tag{3}$$

where I^{pos} (I^{neg}) is an indicator variable which takes a value of 1 if the item is classified as positive (negative) and zero otherwise, and S^{pos} (S^{neg}) is the positive (negative) score of each document or word.

It may be the case that items are classified categorically as either positive or negative rather than scored, so that Equations (1), (2) and (3) simply reduce to the fraction of positive items, the fraction of negative items, and the difference between them, respectively. For formation periods exceeding one day, these measures may either be computed daily before being averaged over the entire formation period (Uhl, 2014; Sinha, 2016), or pooled together at once (Cahan *et al.*, 2017). It is also common for each document to be weighted by some measure of relative importance such as story relevance (Ho *et al.*, 2013; Shi *et al.*, 2016), or, in the case of aggregations over sector and market indices, by firm market capitalisation (Smales, 2015b).

The commonly used measures described above are invariant to any temporal aspects of the news flow over the aggregation period, as information pertaining to the order, dispersion and recency of the news stories is lost through averaging. This is likely to be inconsequential for the daily formation periods typically studied, but may carry more weight over longer formation periods. In this sense, news aggregation techniques may be thought of in terms of a bias-variance trade-off. Daily measures of news flow tend to be noisy with respect to

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low-frequency formation periods and temporal aggregation can assist in extracting the underlying signal of interest (Uhl *et al*., 2015). However, for formation periods spanning multiple months, it would be surprising if the temporal characteristics of the news flow, such as the relative recency of periods of particularly good or bad news exposure, had no bearing on market behaviour.

Vector autoregressive (VAR) models with multiple lags and joint significance tests go some way to alleviate aggregation bias, in that they reduce the temporal invariance from the entire formation period to the length of each lag and capture the linear dynamics between lags. Other studies explicitly use changes in temporal characteristics, such as sentiment spikes, momentum and trend changes, as news events (Uhl *et al*., 2015; Yang *et al*., 2015; Nooijen and Broda, 2016).

The sentiment measures presented above do not account for the total number of stories being considered in each aggregate, which has been shown to be important for both return, volume and volatility estimates (Mitchell and Mulherin, 1994; Tetlock, 2010; Ferguson *et al*., 2015). This information (referred to as news volume, attention or coverage) is usually incorporated into analyses as a separate variable or by weighting the sentiment measures defined above. When measured as a separate variable, the joint effect of news volume with other components of news (such as tone) can be examined explicitly (with the exception of nonparametric models) by including relevant regression terms or conditioning assets on their relative news volume either before or after conditioning on other news variables.

When incorporating news volume into measures of sentiment, different approaches will provide more or less weight to the item count. Antweiler and Frank (2004) discuss this issue when constructing their measure of 'bullishness' from online message boards and present three alternative specifications. The first takes into account the ratio of positive messages minus the ratio of negative messages, but disregards the total number of messages. This is equivalent to *Polarity* as defined above. Their second measure considers the log of the ratio of positive to negative messages and is approximately equal to *Polarity* scaled by the log of the total number of messages. The third measure is the number of positive messages minus the number of messages. This is equal to *Polarity* scaled by the total number of messages and is more sensitive to the total number of messages than the second measure. Antweiler and Frank found that although their key results could be produced using each of the specifications, the performance of each measure was different.

Some studies also orthogonalise news scores to firm characteristics using a regression model to attenuate the extent to which the news scores are proxies for firm size, sector, index membership or other potential determinants of news exposure (Hillert *et al*., 2014; Huynh and Smith, 2017).

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5. Empirical findings

The historiograph illustrated in Figure 1 shows a chronological mapping of the 20 most internally cited review items, generated algorithmically as suggested by Garfield *et al.* (2003) and Garfield (2004). Using the historiographic relationships as a guide, we distil the empirical literature into a number of central themes.

Early work relating news content to stock market behaviour (Niederhoffer, 1971; Wüthrich *et al.*, 1998) focused on the identification of explicit economic events within the text. It was not until Tetlock (2007) that the non-informational effects were examined, that is the influence of news media on investor sentiment. Importantly, Tetlock (2007) demonstrated that even simplistic and imprecise quantifications of news content, such as the frequency of negatively associated words, may contain insightful and even predictive information about market behaviour. Tetlock's (2007) general result has since been repeated and extended with longer sample periods (Garcia, 2013), stricter constraints on market exogeneity (Dougal *et al.*, 2012), and across different markets (Narayan and Bannigidadmath, 2017). This subset of literature examines the relationship between relatively uninformative, market-wide news content and market behaviour.

Tetlock *et al.* (2008) applied the general principles used by Tetlock (2007) to investigate the qualitative information content of news at the firm level. Their results supported the hypothesis that firm-specific news text contains information relevant to fundamentals above and beyond those captured by traditional quantitative measures. Subsequently, similar analyses were performed in different markets (Ferguson *et al.*, 2015), across different types of investor groups (Tetlock, 2011; Hendershott *et al.*, 2015), and with different empirical emphasis (Leinweber and Sisk, 2011; Ahmad *et al.*, 2016). This subset of the literature examines the relationship between informative, firm-specific news content and firm-level market responses.

The market-level effects of uninformative news and firm-level effects of informative news was followed by an amalgamation of these two approaches: the cross-sectional aggregation of firm-specific news content into market- and sector-level news indices (Smales, 2015b). This subset of the literature examines the response to informative news content over different collections of assets, primarily at the market level.

Perhaps due to the empirical specifications chosen in the seminal works of Tetlock (2007) and Tetlock *et al.* (2008), and the significant but short-lived pricing effects they documented, the majority of the literature has stayed within the confines of intra-week analysis. Here, news is typically treated as a daily or per-item variable, which may have a lagged, though discrete, relationship with market behaviour.

Despite the theoretical and empirical associations of investor sentiment and imperfect information processing with gradual pricing effects (DeLong *et al.*,

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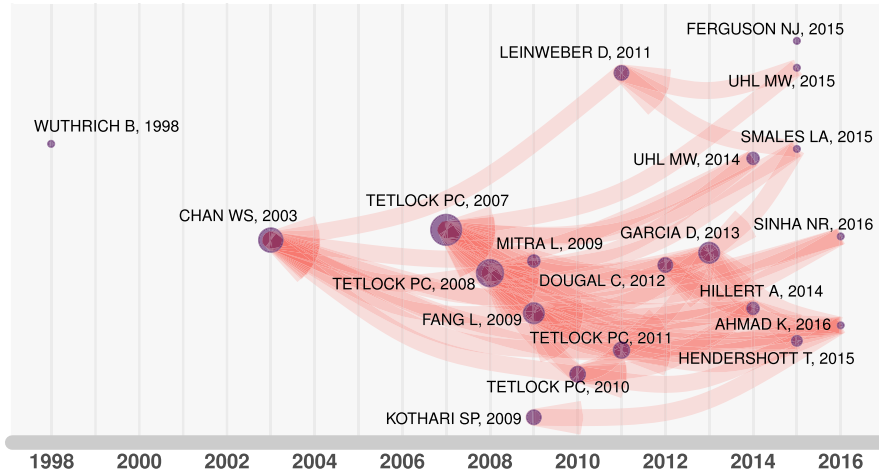

Figure 1 Historiograph of the 20 most internally cited survey items. Figure generated in R using the Bibliometrix package (Aria and Cuccurullo, 2017). [Colour figure can be viewed at wileyonlinelibrary.com]

1990; Daniel *et al*., 1998; Hong and Stein, 1999; Baker and Wurgler, 2006, 2007; Schmeling, 2009), it was not until relatively late in the literature that the role of news analytics in low-frequency market dynamics was explored. By using temporal aggregations of news content, researchers have been able to take advantage of slower-moving news signals and longer forward estimation horizons than those initially examined by Uhl (2014), Uhl *et al*. (2015), and Sinha (2016). As with the intra-week literature, low-frequency literature can further be grouped into investigations at the market or index level, and those at the firm level.

A prominent subset of the low-frequency literature explores the implications of cross-sectional differences in firms' news content through portfolio analyses (see Section 4.2). A major theme among these studies is the interaction between news exposure and stock price momentum (Hillert *et al*., 2014). The investigations of Chan (2003) and Fang and Peress (2009), which made use of news incidence and news coverage, but not news content, have been particularly influential in this corner of the literature.

In both the intra-week and low-frequency literature, a number of researchers have investigated the relationship between news and measures of risk (Kothari *et al*., 2009; Mitra *et al*., 2009). Although this subset of the literature is small, news has been related to realised volatility, idiosyncratic volatility, market implied volatility and tail co-dependence.

Tables 5 and 6 link each of the survey items with the theme of the literature to which they belong, and the key results distilled from this literature are listed below. The specific findings of each study are summarised in the Appendix.

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Table 5
Thematic grouping of intra-week analyses

Publications	Themes
Wüthrich *et al.* (1998) Nuij *et al.* (2014)	Return prediction using automated identification of specific news events
Tetlock (2007) Garcia (2013) Dougal *et al.* (2012) Narayan and Bannigidadmath (2017)	Non-informational media tone, market return and market trading volume
Tetlock *et al.* (2008) Leinweber and Sisk (2011) Tetlock (2011) Ferguson *et al.* (2015) Zhang *et al.* (2016b) Khuu *et al.* (2016) Chan and Chong (2017) Strauß *et al.* (2016)	Firm-level news, return and future earnings. Firm-level news-driven trading strategies. Relative impact of novel versus stale news
Ahmad *et al.* (2016) Garcia (2013) Smales (2015b)	Time variance of news-return relationship at the firm, market and sector level
Allen *et al.* (2017) Yang *et al.* (2017)	Aggregate news tone and market return
Hendershott *et al.* (2015) Strauß *et al.* (2016)	Endogeneity of news
Smales (2015b) Borovkova and Mahakena (2015) Yu *et al.* (2013) Mitra *et al.* (2009) Zhang *et al.* (2016a) Nooijen and Broda (2016)	Asset-level news and volatility
Bianconi *et al.* (2015) Smales (2016)	News and market implied volatility
Zhang *et al.* (2016a,b) Strauß *et al.* (2016) Ferguson *et al.* (2015) Khuu *et al.* (2016) Wüthrich *et al.* (1998) Chan and Chong (2017)	International markets

- Uninformative news tone partly explains subsequent market return and trading volume at intra-week horizons. The price response is characterised by an initial overreaction and subsequent reversal.
- Informative (firm- or asset-specific) news tone partly explains subsequent asset returns at intra-week horizons. The price response is characterised by an initial underreaction and subsequent drift towards equilibrium.
- The short-term price response to informative news is a function of its textual similarity to other recent news items.

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Table 6
Thematic grouping of low-frequency analyses

Publications	Themes
Uhl (2014)	Aggregate news indices and market return.
Uhl *et al.* (2015)	
Kraussl and Mirgorodskaya (2017)	News-driven market timing trading strategies
Kroujiline *et al.* (2016)	
Ammann *et al.* (2014)	
Chan (2003)	Cross-sectional effects of news volume and content on future returns
Fang and Peress (2009)	
Hillert *et al.* (2014)	
Sinha (2016)	
Heston and Sinha (2017)	
Wei *et al.* (2017)	
Song *et al.* (2017)	
Chan (2003)	News and stock price momentum
Hillert *et al.* (2014)	
Huynh and Smith (2017)	
Cahan *et al.* (2017)	Firm-level news and volatility
Kothari *et al.* (2009)	
Shi *et al.* (2016)	
Wei *et al.* (2017)	News, market volatility and implied market volatility
Yang *et al.* (2015)	
Manela and Moreira (2017)	
Kraussl and Mirgorodskaya (2017)	
Wu and Lin (2017)	International markets
Ammann *et al.* (2014)	
Wang *et al.* (2015)	

- A large proportion of the short-term price response to news occurs prior to the release of the news.
- The price response to news is related to the amount and content (polarity) of news items.
- Cross-sectional aggregates of informative news partly explain subsequent returns at the market level.
- Temporal aggregates of informative news over weekly, monthly and multi-month formation periods are related to subsequent pricing dynamics over 1–12-month horizons at the firm and market level.
- Temporal aggregates of informative news over weekly, monthly and multi-month formation periods are related to stock price continuation (momentum). There is no consensus on the underlying mechanism of the relationship.
- News is related to volatility at the firm and market level, with negative (positive) news associated with higher (lower) levels of volatility.

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- News is related to implied volatility at the market level, such that increases in negative (positive) news are associated with positive (negative) changes in implied volatility.

6. Discussion and future research

Financial news media has long been of interest to researchers and practitioners in finance, economics and accounting. In this paper we have surveyed the research applying text analytics to the study of news media in financial markets – a subfield energised by the historically unprecedented capacity to extract, quantify, categorise and aggregate large sets of unstructured news text. Here, we discuss facets of literature that we believe represent fertile ground for further research.

News media is often credited with playing a significant role in the development of market bubbles (Shiller, 2000), but to the best of our knowledge the attribution is thus far anecdotal. While it has been shown that the influence of individual news items on market activity changes with wider economic conditions (Garcia, 2013) and investor confidence (Smales, 2014), the role of the media in stimulating periods of 'irrational exuberance' has not been quantitatively explored.

The literature has revealed new insights about the relationship between quantitative information and news content, as well as the differential impacts of informational and non-informational forms of content. Yet, there has been little research analysing the interaction of these latter two components of news and their joint effect on market prices. It is unclear whether wider media sentiment has any bearing on the effects of firm-level reporting, and if so, whether it acts to dampen or intensify firm-level effects. In a similar light, there appears to have been relatively little research relating the attributes of news items to their reception on social platforms, and whether the intensity of this reception is a useful feature of news flow when forecasting the subsequent market response.

With few exceptions, the literature has made use of only a small number of news features and applied them in similar ways. Further exploration of existing news-based features may yield new insights into the ways these measures relate to one another and how they are most sensibly interpreted. This could include more deliberate analysis of story evolution and the relative impact of multi-piece news items (see Huynh and Smith, 2017 for a recent example), novel ways to include news topics into aggregate measures, and features of content such as story complexity that might lead to less efficient market responses or greater investor disagreement (Dzielinski and Hasseltoft, 2017). Relatedly, the temporally aggregated news variables commonly used in the high-frequency literature may be less suitable when applied to longer formation periods. Incorporating greater temporal resolution of news content into low-frequency analyses, either

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through novel feature transformation or the integration of machine learning models, may shed light on previously unobserved properties of news-market dynamics.

Relatively few papers examine the relationship between firms' news environment and forward stock price volatility, especially at longer horizons. Of those that consider the link between news and implied volatility, the analysis is conducted at the market level. Additionally, there has been almost no consideration of downside characterisations of risk at the frequencies considered. Investigation directed towards identifying and quantifying the news-driven determinants of stock mid- to long-term price dispersion, including downside attributes, is largely uncharted ground for further study.

Statistically significant news-market relationships have now been documented widely, yet the theoretical and practical implications of the associated literature are still in many ways underdeveloped. More research driven by hypotheses grounded in economic theory or motivated by realistic investment environments will be an important factor in the maturation of this field.

References

Admati, A. R., and P. Pfleiderer, 1988, A theory of intraday patterns: volume and price variability, *The Review of Financial Studies* 1, 3–40.

Aggarwal, C. C., and C. Zhai (2012). A survey of text classification algorithms, in: C. C. Aggarwal, C. Zhai, eds., *Mining Text Data* (Springer, New York), 163–222.

Ahmad, K., J. Han, E. Hutson, C. Kearney, and S. Liu, 2016, Media-expressed negative tone and firm-level stock returns, *Journal of Corporate Finance* 37, 152–172.

Akerlof, G. A., 1970, The market for lemons: quality uncertainty and the market mechanism, *Quarterly Journal of Economics* 84, 488–500.

Allen, D. E., M. McAleer, and A. K. Singh, 2017, An entropy-based analysis of the relationship between the DOW JONES Index and the TRNA Sentiment series, *Applied Economics* 49, 677–692.

Ammann, M., R. Frey, and M. Verhofen, 2014, Do newspaper articles predict aggregate stock returns?, *Journal of Behavioral Finance* 15, 195–213.

Antweiler, W., and M. Z. Frank, 2004, Is all that talk just noise? The information content of internet stock message boards, *The Journal of Finance* 59, 1259–1294.

Aria, M., and C. Cuccurullo, 2017, bibliometrix: an R-tool for comprehensive science mapping analysis, *Journal of Informetrics* 11, 959–975.

Baker, M., and J. Wurgler, 2006, Investor sentiment and the cross-section of stock returns, *The Journal of Finance* 61, 1645–1680.

Baker, M., and J. Wurgler, 2007, Investor sentiment in the stock market, *Journal of Economic Perspectives* 21, 129–151.

Banerjee, S., and I. Kremer, 2010, Disagreement and learning: dynamic patterns of trade, *The Journal of Finance* 65, 1269–1302.

Barberis, N., A. Shleifer, and R. Vishny, 1998, A model of investor sentiment, *Journal of Financial Economics* 49, 163–187.

Bianconi, M., X. Hua, and C. M. Tan, 2015, Determinants of systemic risk and information dissemination, *International Review of Economics and Finance* 38, 352–368.

Black, F., 1986, Noise, *The Journal of Finance* 41, 528–543.

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Bollen, J., H. Mao, and X. Zeng, 2011, Twitter mood predicts the stock market, *Journal of Computational Science* 2, 1–8.

Borovkova, S., and D. Mahakena, 2015, News, volatility and jumps: The case of natural gas futures, *Quantitative Finance* 15, 1217–1242.

Boudoukh, J., R. Feldman, S. Kogan, and M. Richardson, 2013, *Which news moves stock prices? A textual analysis*, Working paper (National Bureau of Economic Research, Cambridge, MA).

Cahan, R. H., S. F. Cahan, T. Lee, and N. H. Nguyen, 2017, Media content, accounting quality, and liquidity volatility, *European Accounting Review* 26, 1–25.

Campbell, J. Y., and A. S. Kyle, 1993, Smart money, noise trading and stock price behaviour, *The Review of Economic Studies* 60, 1–34.

Chan, W. S., 2003, Stock price reaction to news and no-news: drift and reversal after headlines, *Journal of Financial Economics* 70, 223–260.

Chan, S. W., and M. W. Chong, 2017, Sentiment analysis in financial texts, *Decision Support Systems* 94, 53–64.

Checkley, M. S., D. A. Higon, and H. Alles, 2017, The hasty wisdom of the mob: how market sentiment predicts stock market behavior, *Expert Systems with Applications* 77, 256–263.

Clark, P. K., 1973, A subordinated stochastic process model with finite variance for speculative prices, *Econometrica: Journal of the Econometric Society*, 41, 135–155.

Clements, A. E., and N. Todorova, 2016, Information flow, trading activity and commodity futures volatility, *Journal of Futures Markets* 36, 88–104.

Crain, S. P., K. Zhou, S.-H. Yang, and H. Zha, 2012, Dimensionality reduction and topic modelling: from latent semantic indexing to latent Dirichlet allocation and beyond, in: C. C. Aggarwal, C. Zhai, eds., *Mining Text Data* (Springer, New York), 129–161.

Daniel, K., M. Grinblatt, S. Titman, and R. Wermers, 1997, Measuring mutual fund performance with characteristic-based benchmarks, *The Journal of Finance* 52, 1035–1058.

Daniel, K., D. Hirshleifer, and A. Subrahmanyam, 1998, Investor psychology and security market under- and overreactions, *The Journal of Finance* 53, 1839–1885.

Das, S. R., 2014, Text and context: language analytics in finance, *Foundations and Trends in Finance* 8, 145–261.

De Bondt, W. F. M., and R. Thaler, 1985, Does the stock market overreact?, *The Journal of Finance* 40, 793–805.

De Jong, K., 1988, Learning with genetic algorithms: an overview, *Machine Learning* 3, 121–138.

DeLong, J. B., A. Shleifer, L. H. Summers, and R. J. Waldmann, 1990, Noise trader risk in financial markets, *Journal of Political Economy* 98, 703–738.

Dougal, C., J. Engelberg, D. Garcia, and C. A. Parsons, 2012, Journalists and the stock market, *The Review of Financial Studies* 25, 639–679.

Dzielinski, M., and H. Hasseltoft (2017). News tone dispersion and investor disagreement. Available at: https://ssrn.com/abstract=2192532.

Easley, D., and M. O'Hara, 2004, Information and the cost of capital, *The Journal of Finance* 59, 1553–1583.

Engelberg, J. E., and C. A. Parsons, 2011, The causal impact of media in financial markets, *The Journal of Finance* 66, 67–97.

Engle, R. F., M. K. Hansen, and A. Lunde, 2012, And now, the rest of the news: volatility and firm specific news arrival. CREATES Research Paper, Department of Economics and Business Economics, Aarhus University. Available at: https://ideas.repec.org/p/aah/create/2012-56.html.

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Epps, T. W., and M. L. Epps, 1976, The stochastic dependence of security price changes and transaction volumes: implications for the mixture-of-distributions hypothesis, *Econometrica* 44, 305–321.
Fama, E. F., 1998, Market efficiency, long-term returns, and behavioral finance, *Journal of Financial Economics* 49, 283–306.
Fang, L., and J. Peress, 2009, Media coverage and the cross-section of stock returns, *The Journal of Finance* 64, 2023–2052.
Ferguson, N. J., D. Philip, H. Y. T. Lam, and J. M. Guo, 2015, Media content and stock returns: the predictive power of press, *Multinational Finance Journal* 19, 1–31.
Feuerriegel, S., and H. Prendinger, 2016, News-based trading strategies, *Decision Support Systems* 90, 65–74.
Fisher, R. A., 1936, The use of multiple measurements in taxonomic problems, *Annals of Human Genetics* 7, 179–188.
Foster, F. D., and S. Viswanathan, 1990, A theory of the interday variations in volume, variance, and trading costs in securities markets, *The Review of Financial Studies* 3, 593–624.
Foster, F. D., and S. Viswanathan, 1993a, The effect of public information and competition on trading volume and price volatility, *The Review of Financial Studies* 6, 23–56.
Foster, F. D., and S. Viswanathan, 1993b, Variations in trading volume, return volatility, and trading costs: evidence on recent price formation models, *The Journal of Finance* 48, 187–211.
Fung, G. P. C., J. X. Yu, and W. Lam, 2003, Automatic stock trend prediction by real time news, in: W.-K. Ching, M. K.-P. Ng, eds., *Advances in Data Mining and Modelling* (World Scientific, Singapore), 48–59.
Garcia, D., 2013, Sentiment during recessions, *The Journal of Finance* 68, 1267–1300.
Garfield, E., 2004, Historiographic mapping of knowledge domains literature, *Journal of Information Science* 30, 119–145.
Garfield, E., A. I. Pudovkin, and V. S. Istomin, 2003, Why do we need algorithmic historiography?, *Journal of the American Society for Information Science and Technology* 54, 400–412.
Geva, T., and J. Zahavi, 2014, Empirical evaluation of an automated intraday stock recommendation system incorporating both market data and textual news, *Decision Support Systems* 57, 212–223.
Grefenstette, G., and P. Tapanainen, 1994, What is a word, what is a sentence? Problems of tokenisation, The 3rd International Conference on Computational Lexicography, Budapest, 79–87.
Griffin, J. M., N. H. Hirschey, and P. J. Kelly, 2011, How important is the financial media in global markets?, *The Review of Financial Studies* 24, 3941–3992.
Groß-Klußmann, A., and N. Hautsch, 2011, When machines read the news: using automated text analytics to quantify high frequency news-implied market reactions, *Journal of Empirical Finance* 18, 321–340.
Grossman, S. J., and J. E. Stiglitz, 1976, Information and competitive price systems, *American Economic Review* 66, 246–253.
Gutierrez, R. C., and E. K. Kelly, 2008, The long-lasting momentum in weekly returns, *The Journal of Finance* 63, 415–447.
Hagenau, M., M. Liebmann, and D. Neumann, 2013, Automated news reading: stock price prediction based on financial news using context-capturing features, *Decision Support Systems* 55, 685–697.
Hart, R. P., 2000, *DICTION 5.0: The Text Analysis Program* (Digitex, Austin, TX). Available at: https://www.jou.ufl.edu/assets/researchlab/dictionmanual.pdf

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He, H., and J. Wang, 1995, Differential information and dynamic behavior of stock trading volume, *The Review of Financial Studies* 8, 919–972.
Hendershott, T., D. Livdan, and N. Schuerhoff, 2015, Are institutions informed about news?, *Journal of Financial Economics* 117, 249–287.
Henry, E., 2008, Are investors influenced by how earnings press releases are written?, *The Journal of Business Communication (1973)* 45, 363–407.
Heston, S. L., and N. R. Sinha, 2017, News vs. sentiment: predicting stock returns from news stories, *Financial Analysts Journal* 73, 67–83.
Hillert, A., H. Jacobs, and S. Mueller, 2014, Media makes momentum, *The Review of Financial Studies* 27, 3467–3501.
Hirshleifer, D., and S. H. Teoh, 2003, Limited attention, information disclosure, and financial reporting, *Journal of Accounting and Economics* 36, 337–386.
Ho, K.-Y., Y. Shi, and Z. Zhang, 2013, How does news sentiment impact asset volatility? Evidence from long memory and regime-switching approaches, *North American Journal of Economics and Finance* 26, 436–456.
Holden, C. W., and A. Subrahmanyam, 1992, Long-lived private information and imperfect competition, *The Journal of Finance* 47, 247–270.
Hong, H., and J. C. Stein, 1999, A unified theory of underreaction, momentum trading, and overreaction in asset markets, *The Journal of Finance* 54, 2143–2184.
Hu, M., and B. Liu, 2004, Mining and summarizing customer reviews. *Proceedings of the 10th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 168–177.
Huynh, T. D., and D. R. Smith, 2017, Stock price reaction to news: the joint effect of tone and attention on momentum, *Journal of Behavioral Finance* 18, 304–328.
Jegadeesh, N., and S. Titman, 1993, Returns to buying winners and selling losers: implications for stock market efficiency, *The Journal of Finance* 48, 65–91.
Kandel, E., and N. D. Pearson, 1995, Differential interpretation of public signals and trade in speculative markets, *Journal of Political Economy* 103, 831–872.
Kearney, C., and S. Liu, 2014, Textual sentiment in finance: a survey of methods and models, *International Review of Financial Analysis* 33, 171–185.
Khuu, J., R. B. Durand, and L. A. Smales, 2016, Melancholia and Japanese stock returns – 2003 to 2012, *Pacific-Basin Finance Journal* 40, 424–437.
Kothari, S., X. Li, and J. E. Short, 2009, The effect of disclosures by management, analysts, and business press on cost of capital, return volatility, and analyst forecasts: a study using content analysis, *The Accounting Review* 84, 1639–1670.
Kraussl, R., and E. Mirgorodskaya, 2017, Media, sentiment and market performance in the long run, *European Journal of Finance* 23, 1059–1082.
Kroujiline, D., M. Gusev, D. Ushanov, S. V. Sharov, and B. Govorkov, 2016, Forecasting stock market returns over multiple time horizons, *Quantitative Finance* 16, 1695–1712.
Kyle, A. S., 1985, Continuous auctions and insider trading, *Econometrica* 53, 1315–1335.
Lee, S., J. Song, and Y. Kim, 2010, An empirical comparison of four text mining methods, *Journal of Computer Information Systems* 51, 1–10.
Leinweber, D., and J. Sisk, 2011, Event-driven trading and the new news, *The Journal of Portfolio Management* 38, 110–124.
Leung, H., and T. Ton, 2015, The impact of internet stock message boards on cross-sectional returns of small-capitalization stocks, *Journal of Banking and Finance* 55, 37–55.
Li, F., 2010, Textual analysis of corporate disclosures: a survey of the literature, *Journal of Accounting Literature* 29, 143–167.

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Li, Q., T. Wang, P. Li, L. Liu, Q. Gong, and Y. Chen, 2014a, The effect of news and public mood on stock movements, *Information Sciences* 278, 826–840.
Li, X., H. Xie, L. Chen, J. Wang, and X. Deng, 2014b, News impact on stock price return via sentiment analysis, Knowledge-Based Systems 6, 14–23.
Liu, B., and L. Zhang (2012). A survey of opinion mining and sentiment analysis, in: C. C. Aggarwal, C. C. Zhai, eds., *Mining Text Data* (Springer, New York). 415–463.
Loughran, T., and B. McDonald, 2011, When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks, *The Journal of Finance* 66, 35–65.
Loughran, T., and B. McDonald, 2016, Textual analysis in accounting and finance: a survey, *Journal of Accounting Research* 54, 1187–1230.
Lu, Y.-C., and Y.-C. Wei, 2013, The Chinese news sentiment around earnings announcements, *Romanian Journal of Economic Forecasting* 16, 44–58.
Manela, A., and A. Moreira, 2017, News implied volatility and disaster concerns, *Journal of Financial Economics* 123, 137–162.
Maragoudakis, M., and D. Serpanos, 2016, Exploiting financial news and social media opinions for stock market analysis using MCMC Bayesian inference, *Computational Economics* 47, 589–622.
Maslyuk-Escobedo, S., K. Rotaru, and A. Dokumentov, 2017, News sentiment and jumps in energy spot and futures markets, *Pacific-Basin Finance Journal* 45, 186–210.
Merton, R. C., 1987, A simple model of capital market equilibrium with incomplete information, *The Journal of Finance* 42, 483–510.
Milgrom, P., and N. Stokey, 1982, Information, trade and common knowledge, *Journal of Economic Theory* 26, 17–27.
Mitchell, M. L., and J. H. Mulherin, 1994, The impact of public information on the stock market, *The Journal of Finance* 49, 923–950.
Mitra, G., and L. Mitra, 2011, *The Handbook of News Analytics in Finance* (John Wiley & Sons, Chichester).
Mitra, L., G. Mitra, and D. Dibartolomeo, 2009, Equity portfolio risk estimation using market information and sentiment, *Quantitative Finance* 9, 887–895.
Narayan, P. K., and D. Bannigidadmath, 2017, Does financial news predict stock returns? New evidence from Islamic and non-Islamic stocks, *Pacific-Basin Finance Journal* 42, 24–45.
Nardo, M., M. Petracco-Giudici, and M. Naltsidis, 2016, Walking down Wall Street with a tablet: a survey of stock market predictions using the web, *Journal of Economic Surveys* 30, 356–369.
Nassirtoussi, A. K., S. Aghabozorgi, T. Y. Wah, and D. C. L. Ngo, 2014, Text mining for market prediction: a systematic review, *Expert Systems with Applications* 41, 7653–7670.
Nassirtoussi, A. K., S. Aghabozorgi, T. Y. Wah, and D. C. L. Ngo, 2015, Text mining of news-headlines for FOREX market prediction: a multi-layer dimension reduction algorithm with semantics and sentiment, *Expert Systems with Applications* 42, 306–324.
Newey, W. K., and K. D. West, 1987, A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix, *Econometrica* 55, 703–708.
Niederhoffer, V., 1971, The analysis of world events and stock prices, *The Journal of Business* 44, 193–219.
Nooijen, S. J., and S. A. Broda, 2016, Predicting equity markets with digital online media sentiment: evidence from Markov-switching models, *Journal of Behavioral Finance* 17, 321–335.
Nuij, W., V. Milea, F. Hogenboom, F. Frasincar, and U. Kaymak, 2014, An automated framework for incorporating news into stock trading strategies, *IEEE Transactions on Knowledge and Data Engineering* 26, 823–835.

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Peng, L., and W. Xiong, 2006, Investor attention, overconfidence and category learning, *Journal of Financial Economics* 80, 563–602.
Ravi, K., and V. Ravi, 2015, A survey on opinion mining and sentiment analysis: tasks, approaches and applications, *Knowledge-Based Systems* 89, 14–46.
Schmeling, M., 2009, Investor sentiment and stock returns: some international evidence, *Journal of Empirical Finance* 16, 394–408.
Schumaker, R. P., and H. Chen, 2009, Textual analysis of stock market prediction using breaking financial news: the AZFin text system, *ACM Transactions on Information Systems (TOIS)* 27, 12.
Schumaker, R. P., Y. Zhang, C.-N. Huang, and H. Chen, 2012, Evaluating sentiment in financial news articles, *Decision Support Systems* 53, 458–464.
Sebastiani, F., 2002, Machine learning in automated text categorization, *ACM Computing Surveys* 34, 1–47.
Seng, J.-L., and H.-F. Yang, 2017, The association between stock price volatility and financial news – a sentiment analysis approach, *Kybernetes* 46, 1341–1365.
Shi, Y., K.-Y. Ho, and W.-M. Liu, 2016, Public news arrival and the idiosyncratic volatility puzzle, *Journal of Empirical Finance* 37, 159–172.
Shiller, R. C., 2000, Irrational exuberance, *Philosophy and Public Policy Quarterly* 20, 18–23.
Sinha, N. R., 2016, Underreaction to news in the US stock market, *Quarterly Journal of Finance* 6, 1650005.
Smales, L. A., 2014, News sentiment and the investor fear gauge, *Finance Research Letters* 11, 122–130.
Smales, L. A., 2015a, Asymmetric volatility response to news sentiment in gold futures, *Journal of International Financial Markets, Institutions and Money* 34, 161–172.
Smales, L. A., 2015b, Time-variation in the impact of news sentiment, *International Review of Financial Analysis* 37, 40–50.
Smales, L. A., 2016, Time-varying relationship of news sentiment, implied volatility and stock returns, *Applied Economics* 48, 4942–4960.
Song, Q., A. Liu, and S. Y. Yang, 2017, Stock portfolio selection using learning-to-rank algorithms with news sentiment, *Neurocomputing* 264, 20–28.
Stanford NLP Group, 2018, *Stanford CoreNLP – Natural language software*. Available at: https://stanfordnlp.github.io/CoreNLP/.
Stone, P. J., D. C. Dunphy, and M. S. Smith, 1966. The general inquirer: a computer approach to content analysis (MIT Press, Cambridge, MA).
Strauß, N., R. Vliegenthart, and P. Verhoeven, 2016, Lagging behind? Emotions in newspaper articles and stock market prices in the Netherlands, *Public Relations Review* 42, 548–555.
Tauchen, G. E., and M. Pitts, 1983, The price variability-volume relationship on speculative markets, *Econometrica* 51, 485–505.
Tetlock, P. C., 2007, Giving content to investor sentiment: the role of media in the stock market, *The Journal of Finance* 62, 1139–1168.
Tetlock, P. C., 2010, Does public financial news resolve asymmetric information?, *The Review of Financial Studies* 23, 3520–3557.
Tetlock, P. C., 2011, All the news that's fit to reprint: do investors react to stale information?, *The Review of Financial Studies* 24, 1481–1512.
Tetlock, P. C., 2014, Information transmission in finance, *Annual Review of Financial Economics* 6, 365–384.
Tetlock, P. C., M. Saar-Tsechansky, and S. Macskassy, 2008, More than words: quantifying language to measure firms' fundamentals, *The Journal of Finance* 63, 1437–1467.

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Thomson Reuters, 2013, *Thomson Reuters News Analytics: Sentiment analysis, relevance, novelty*. White paper (Thomson Reuters).

Uhl, M. W., 2014, Reuters sentiment and stock returns, *Journal of Behavioral Finance* 15, 287–298.

Uhl, M. W., M. Pedersen, and O. Malitius, 2015, What's in the news? Using news sentiment momentum for tactical asset allocation, *Journal of Portfolio Management* 41, 100–112.

Wang, J., 1994, A model of competitive stock trading volume, *Journal of Political Economy* 102, 127–168.

Wang, K.-Y., C.-K. Chen, and H.-C. Wei, 2015, Media coverage and stock return in the Taiwan stock market, *Acta Oeconomica* 65, 35–53.

Wei, Y.-C., Y.-C. Lu, J.-N. Chen, and Y.-J. Hsu, 2017, Informativeness of the market news sentiment in the Taiwan stock market, *North American Journal of Economics and Finance* 39, 158–181.

White, H., 1980, A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity, *Econometrica* 48, 817–838.

Wilson, T., J. Wiebe, and P. Hoffmann, 2005, Recognizing contextual polarity in phrase-level sentiment analysis, *Proceedings of HLT-EMNLP-2005*, 347–354.

Wu, C.-H., and C.-J. Lin, 2017, The impact of media coverage on investor trading behavior and stock returns, *Pacific-Basin Finance Journal* 43, 151–172.

Wüthrich, B., D. Permunetilleke, S. Leung, W. Lam, V. Cho, and J. Zhang, 1998, Daily prediction of major stock indices from textual www data, *HKIE Transactions* 5, 151–156.

Xiong, G., and S. Bharadwaj, 2013, Asymmetric roles of advertising and marketing capability in financial returns to news: turning bad into good and good into great, *Journal of Marketing Research* 50, 706–724.

Yang, S. Y., Q. Song, S. Y. K. Mo, K. Datta, and A. Deane, 2015, The impact of abnormal news sentiment on financial markets, *Journal of Business and Economics* 6, 1682–1694.

Yang, S. Y., S. Y. K. Mo, A. Liu, and A. A. Kirilenko, 2017, Genetic programming optimization for a sentiment feedback strength based trading strategy, *Neurocomputing* 264, 29–41.

Yu, Y., W. Duan, and Q. Cao, 2013, The impact of social and conventional media on firm equity value: a sentiment analysis approach, *Decision Support Systems* 55, 919–926.

Zhang, J. L., W. K. Härdle, C. Y. Chen, and E. Bommes, 2016a, Distillation of news flow into analysis of stock reactions, *Journal of Business and Economic Statistics* 34, 547–563.

Zhang, Y., W. Song, D. Shen, and W. Zhang, 2016b, Market reaction to internet news: information diffusion and price pressure, *Economic Modelling* 56, 43–49.

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Appendix Summary of survey items

Author	Assets	Timeframe	Findings
Wüthrich *et al.* (1998)	DJIA, Nikkei 225, FT100, Hang Seng, Singapore Straits	6 December 1997–6 March 1998	The occurrence of predefined words and phrases in news headlines can help predict direction of next-day market returns through statistical learning techniques. A probabilistic rule-based classifier was found to outperform rival techniques
Chan (2003)	Random 25% of CRSP stocks	1980–2000	Stocks with extreme monthly returns accompanied by news exhibit stock price momentum for the following 12 months, while those without news tend to reverse the subsequent month; the effect is focused among small stocks with negative returns; excluding no-news stocks from a standard momentum portfolio can significantly improve performance
Tetlock (2007)	DJIA	1980–2008	High values of media pessimism are associated with lower same-day and next-day stock returns, followed by a reversion to fundamentals within one week; unusually high or low values of media pessimism lead to temporarily high market trading volume; media pessimism weakly predicts market volatility; market responses to media pessimism are unlikely to be due to information content
Tetlock *et al.* (2008)	S&P 500 constituents	1980–2004	The fraction of negative words in firm-specific news stories explains future earnings above and beyond stock analysts' forecasts and accounting data; stock prices respond to negative words in news items with a one-day delay; negative words in news stories about earnings are more predictive of earnings and returns than negative words in other news stories

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Appendix (continued)

Author	Assets	Timeframe	Findings
Mitra *et al.* (2009)	Selected constituents of EUROSTOXX 50 and Dow Jones 30	17–23 January 2008; 18–24 September 2008	Incorporation of news sentiment into volatility models can increase the responsiveness of volatility estimates to market developments
Kothari *et al.* (2009)	889 US listed firms	1996–2001	Positive and negative news media impacts cost of capital, return volatility and analyst forecast dispersion; negative disclosures from management lead to higher return volatility and analyst forecast dispersion; analysts' reports and positive disclosures from management appear to be heavily discounted by the market
Fang and Peress (2009)	All firms listed on NYSE + 500 randomly selected firms listed on NASDAQ	1993–2002	Stocks with no media coverage earn higher returns than stocks with high media coverage after controlling for well-known risk factors, including the probability of informed trading; a profitable zero-cost portfolio can be constructed by going long no-coverage stocks and short high-coverage stocks; the media coverage effect is largest among small, low-priced stocks
Tetlock (2010)	NYSE, AMEX and NASDAQ common stocks	1979–2007	Ten-day return reversals are significantly lower on news days; 10-day volume-induced momentum exists only on news days for small and illiquid stocks; news stories with the most informative characteristics are the most powerful predictors of lower return reversal; results consistent with model in which news resolves asymmetrically held information
Leinweber and Sisk (2011)	S&P 1500 constituents	2004–2008; Jan-Sep 2010	Stock prices respond to the tone of firm-specific news with some delay; post-news drift is more pronounced for novel news items with extreme tone; post announcement drift varies by firm market capitalisation and industry; going long (short) stocks with extremely positive (negative) news can form the basis of a profitable trading strategy

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Appendix (continued)

Author	Assets	Timeframe	Findings
Tetlock (2011)	10,187 US firms	1996–2008	Stock prices respond to stale news less than novel news; 1 week return reversal increases with the staleness of the news; the impact of news staleness on return reversal is much larger in stocks with above-median individual ownership; individual investors trade more aggressively on stale news
Dougal *et al*. (2012)	DJIA	1970–2007	WSJ columnists exhibit measurable differences in writing style; unconditional and conditional columnist fixed effects have significant explanatory power for next-day market returns
Garcia (2013)	DJIA	1905–2005	Media tone helps predict market returns at the daily frequency, and this predictability is greater during recessions; the effect of media tone on next-day returns partially reverses over the following 4 days; news tone leads to an increase in trading volume
Xiong and Bharadwaj (2013)	141 large US firms	2004–2010	Positive (negative) news sentiment is associated with higher (lower) abnormal returns at a monthly frequency. Firm advertising amplifies the impact of positive news on stock price but does not affect the impact of negative news on stock price
Yu *et al*. (2013)	824 randomly selected US firms	July–September 2011	Social media appears to have a stronger effect on stock price than news media; social media and news media have strong interaction effects on stock price; both social media and news media appear to be more strongly related to stock price volatility than abnormal returns
Hillert *et al*. (2014)	All common stocks traded on NYSE, AMEX or NASDAQ subject to price and size constraints	1989–2010	Stock price momentum increases with firm media coverage; the stock price momentum of high-coverage firms is stronger for firms with concordant news tone; the momentum of high-coverage firms reverses in the long run; the momentum of high-coverage firms appears to be the result of investor overreaction

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Appendix (continued)

Author	Assets	Timeframe	Findings
Uhl (2014)	DJIA	2003–2010	Changes in aggregate news tone can predict and explain changes in monthly stock market returns; negative news tone has greater predictive capacity than positive news tone; aggregate news tone explains and predicts changes in stock market returns more effectively than market trading volume and certain macroeconomic factors; a trading strategy based on market forecasts which incorporate aggregate news tone can significantly outperform the market
Ammann *et al.* (2014)	DAX	1989–2011	Word count indices of financial news can help predict monthly stock market returns; the predictive capacity of specific words contained in news varies through time; the overall prediction accuracy of word count indices varies through time and has increased since 2000; clustering of word count indices increases the forecast accuracy of estimates relative to those made with raw word counts
Li *et al.* (2014b)	22 Hang Seng Index constituents	2003–2008	In predicting the direction of daily stock movements using support vector machines: the use of sentiment dictionaries outperformed using term frequencies alone, using multiple dimensions (word categories) of dictionaries outperformed using only positive and negative categories, LM slightly outperformed H-IV4
Nuij *et al.* (2014)	FTSE 350 constituents	January–April 2007	Event-based variables extracted from news items are often included in optimal trading rules selected among technical indicators by genetic algorithms based on strategy performance; authors present framework for automatically incorporating news into stock trading strategies

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Appendix (continued)

Author	Assets	Timeframe	Findings
Smales (2015b)	12 US industry indices	2004–2010	The strength of the relationship between industry-specific news tone and stock returns varies through time and by industry and is related to the market risk premium; during periods of high investor fear, the relationship between industry-specific news and stock returns is weak
Hendershott *et al.* (2015)	1667 firms listed on NYSE stock exchange	2003–2005	Institutional order volume predicts the occurrence of news stories; institutional order flow predicts the tone and stock market reaction of next-day news items; Institutional order flow predicts event-day returns for unexpected, value-destroying news events; institutions do not appear to be responsible for overreactions following hype news
Ferguson *et al.* (2015)	FTSE 100 constituents	1981–2010	Positive and negative news tone predict next-day stock returns; news volume is a stronger predictor of next-day stock returns than news tone; the predictive power of news tone is stronger among smaller firms and those with high book-to-market ratios; a trading strategy based on firms' average daily news tone generates positive excess returns over the sample period
Wang *et al.* (2015)	TSEC-listed (Taiwanese) companies	2001–2012	Daily news coverage and tone are related to future daily stock returns for up to 5 days; news coverage and amount are autocorrelated; daily news coverage is related to past stock returns for up to 5 days
Yang *et al.* (2015)	SPY	2012–2014	Shocks and trends in aggregate news tone predict stock market returns and implied volatility over a number of days; news tone trends are stronger predictors of stock market movements than news tone shocks; a trading strategy based on the occurrence of news tone shocks and trends outperforms the market over the sample period

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Appendix (continued)

Author	Assets	Timeframe	Findings
Smales (2015a)	Gold futures	2003–2012	The tone of macroeconomic news is significantly related to the volatility of gold futures; the volatility response of gold futures to news is much stronger for negative news than positive news; the volatility response to news increased dramatically during the GFC recessionary period
Bianconi *et al.* (2015)	14 large US financial firms	1992–2006	Media pessimism impacts financial firms' tail risk through tail covariance of other firms' risk (externality); VIX has a direct effect on systematic risk and via other firms' financial distress; media pessimism and VIX exhibit dynamic feedback at 1–2 day horizons; VIX appears to have greater predictive capacity than media pessimism when forecasting systematic risk
Borovkova and Mahakena (2015)	Natural gas futures	2006–2010	News tone Granger-causes price jumps; news tone is Granger-caused by volatility (especially downside semi-volatility); arrival of news in non-trading periods impacts overnight returns; including news tone in volatility models significantly improves the accuracy of volatility forecasts
Ahmad *et al.* (2016)	20 large non-financial US firms	2001–2010	The prediction strength of negative news tone on stock returns at the firm level passes through periods of significance and non-significance; each episode of significance may be characterised by transitory (short-term reversal) or lasting responses to news tone; market response to newswires appears to be more permanent than that of other types of news; price response to news is most pronounced at the 1-day horizon

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Appendix (continued)

Author	Assets	Timeframe	Findings
Zhang *et al.* (2016a)	100 S&P constituents	2009–2014	Positive and negative news tone impacts returns, volume and volatility at 1–2 day horizons; the strength of the relationship between news tone and market response varies by sector and firm media coverage; the market response to news tone is asymmetric
Strauß *et al.* (2016)	21 firms listed on Amsterdam exchange	2002–2013	Neither the tone nor volume of news stories demonstrate consistent effects on next-day opening prices; news tone appears to be influenced by changes in opening prices rather than the other way around
Zhang *et al.* (2016b)	242 Chinese stocks	November 2013–January 2014	Market responses to internet news differ depending on the type of the news (market commentary versus firm recommendations); firms mentioned in internet news experience abnormal returns and excess trading volume the day of the news
Khuu *et al.* (2016)	Firms traded on Tokyo stock exchange	2003–2012	Negative news sentiment is associated with negative returns at the firm and market level; the effect of news sentiment is largest among smaller firms; news sentiment for Japan-listed firms was predominantly negative over the sample period
Sinha (2016)	US common stocks	2003–2010	Weekly news sentiment predicts stock returns for 13 weeks; drift following weekly news sentiment is more pronounced for negative news; a portfolio long (short) stocks with highly positive (negative) news tone generates positive abnormal returns

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Appendix (continued)

Author	Assets	Timeframe	Findings
Kroujiline *et al.* (2016)	SPY	2003–2016	A news-driven, analytic, agent-based market model with heterogenous investors is able to capture stock market price dynamics at various horizons; the stock market exhibits efficiency to news at intraday timescales but becomes inefficient at longer horizons; a trading strategy based on the model's forecasts, using measured news tone as an input, generates significant abnormal returns
Nooijen and Broda (2016)	MSCI US Equity Sector Indices	1998–2013	News and social media tone more strongly predict volatility than returns; news tone is a better predictor of volatility than social media tone; stock price sensitivity to news tone varies by sector; news and social media tone appear to have a larger impact on prices during stressed (high volatility) markets
Smales (2016)	S&P 500 Index	2000–2010	Changes in VIX are significantly negatively related to changes in news sentiment; the VIX-tone relationship is stronger during periods of high volatility; current and lagged news tone is significantly positively related to stock returns and significantly negatively related to VIX; VIX appears better at forecasting future returns than news tone
Shi *et al.* (2016)	NYSE, AMEX and NASDAQ common stocks	2000–2011	The idiosyncratic volatility puzzle (negative risk premium) appears to be due to the confounding relationship between news (volume and tone), abnormal returns and expected volatility; stock return is negatively (positively) related to the volume of negative (positive) news; idiosyncratic volatility is positively (negatively) related to the volume of negative (positive) news

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Appendix (continued)

Author	Assets	Timeframe	Findings
Allen *et al.* (2017)	DJIA	2007–2012	Aggregate news tone is positively related to market returns; the news tone–return relationship was significantly stronger during the GFC; the entropy of news tone is greater than the entropy of market returns
Yang *et al.* (2017)	SPY	August 2012–January 2015	Sentiment feedback indicators extracted from news and social media are recognised as optimal trading rules (among common technical indicators) by genetic algorithm selection based on strategy performance; sentiment feedback indicators may help reduce the maximum drawdown of algorithmic trading strategies
Kraussl and Mirgorodskaya (2017)	MSCI World and S&P 500 Indices	1990–2012	Media pessimism Granger-causes global market returns for 12–24 months in advance and volatility for 1–24 months in advance; media pessimism is associated with positive (negative) market returns 24–25 (14–17) months in advance and positive market volatility 1–20 months in advance
Wei *et al.* (2017)	All firms on Taiwan Stock Exchange	2003–2012	Aggregate news tone appears to be related to market behaviour only within a particular range of news tone. News tone is related to stock returns at the firm level only for a subset of stocks with particular characteristics
Heston and Sinha (2017)	US common stocks	2003–2010	Weekly news tone predicts stock returns for 13 weeks, whereas daily news tone predicts stock returns for only 1–2 days; drift following weekly news tone is concentrated on negative news; drift following weekly news tone is slightly more pronounced for firms with multiple news items; a portfolio long (short) stocks with highly positive (negative) news tone generates positive abnormal returns and is highly correlated with 26-week momentum and earnings surprise decile spreads

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Appendix (continued)

Author	Assets	Timeframe	Findings
Song *et al*. (2017)	512 heavily traded US stocks	2003–2014	News tone, including news shock and trend indicators, can be incorporated into relative performance forecasts to produce reliable rankings of the best and worst performing stocks at 1-week horizons; these news-informed rankings can be incorporated into trading strategies that generate positive excess returns over the sample period
Huynh and Smith (2017)	Intersection of TRNA and TRTH database	2003–2011	Market underreaction to positive news is a driver of weekly momentum returns (as in Gutierrez and Kelly, 2008); underreaction is greater for stories with high journalist attention; a trading strategy formed on the basis of news tone and past returns is no longer profitable after high-attention news is controlled for; a strategy buying (selling) winners (losers) with high-attention (low attention) positive (negative) news is highly profitable in the US and 21 developed markets
Cahan *et al*. (2017)	2683 NYSE- and AMEX-listed stocks	2003–2011	Contemporaneous and lagged news tone is negatively associated with return volatility over monthly horizons; positive and negative news tone reduces liquidity volatility; the liquidity volatility–news tone relationship is conditional on the firm's accounting quality
Seng and Yang (2017)	Companies listed on Taiwanese stock exchange excluding F-stocks and TDRs	2013–2014	News tone is associated with future stock return volatility at monthly, quarterly, semi-annual and annual horizons
Chan and Chong (2017)	Hang Seng Index	September 2014–March 2015	Tone extracted from online news articles and blogs is associated with market returns at 3- and 5-day horizons. SentiWordNet outperforms H-IV4 dictionary in sentiment classification for a range of models. Grammar-based models outperform word-based models in sentiment classification

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Appendix (continued)

Author	Assets	Timeframe	Findings
Manela and Moreira (2017)	S&P 500 Index	1890–2009	Front page macroeconomic and political news captures investor uncertainty. Time variation in this uncertainty drives expected returns at horizons of 6–24 months. News implied volatility is associated with disaster regime transition probability
Narayan and Bannigidadmath (2017)	18 Dow Jones stock indices	1996–2012	News tone predicts stock returns across Asia-Pacific, Emerging Markets, Japan, Canada and UK, but not the US. News tone more accurately predicts Islamic stocks than conventional stocks
Wu and Lin (2017)	Taiwanese stocks (TSE)	2001–2014	Foreign institutional investment is associated with the quantity and content of news. News tone is contemporaneously associated with abnormal returns

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